

Polynomial divisibility of height-one factorial ratios

Proofs of Bala’s uniform-multiplier conjectures,
a complete rational-root criterion,
and certified optimal constants

A research attempt with reproducible exact certificates
September 19, 2026

Main outcome

For

$$A(n) = \frac{(30n)! n!}{(15n)! (10n)! (6n)!},$$

we prove the all- r divisibility conjectures still recorded under OEIS A211417. We also classify every polynomial splitting over $\mathbb{Q}$ that can uniformly divide $A(n)$ after multiplication by one fixed integer. The analogous conjectures for OEIS A295431 follow from the same theorem.

Abstract. Let $R(n)$ be a balanced integral factorial ratio and let $P \in \mathbb{Z}[X]$ split over $\mathbb{Q}$. We characterize the existence of a nonzero integer C with $CR(n)/P(n) \in \mathbb{Z}$ for every nonnegative integer n at which the quotient is defined. The criterion compares the multiplicity of each rational root of P with one-sided values of the Landau step function. We prove both directions and give an explicit multiplier. At height one, the criterion reduces to the signs of cyclotomic exponents and excludes all repeated roots. Applications settle uniform-product conjectures of Peter Bala, including explicit all- r bounds. Finite weighted digit graphs determine the eight particular proposed constants and prove each optimal. The associated integer sequences admit explicit recurrences, hypergeometric generating functions, and all-order asymptotic expansions. The archive contains exact certificates, a separate verifier, regression tests, and an admissible-denominator atlas for all 52 sporadic examples.

Part II (October 1, 2026, batch 74): integrality and algebraicity for OEIS A347854–A347858 by rational dilation, and the divisor $6(6n+1)(12n-1)$ of A295432; its A295431 multipliers re-prove Part I’s.

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Part I. Polynomial divisibility of height-one factorial ratios

1 The selected problem and its status

1.1 Two exceptional integer sequences

Consider the factorial ratios

$$A(n) = \frac{(30n)!n!}{(15n)!(10n)!(6n)!}, \quad B(n) = \frac{(12n)!n!}{(6n)!(4n)!(3n)!}, \quad n \geq 0. \quad (1)$$

They are OEIS A211417 and A295431, respectively. Both belong to the 52 sporadic height-one integral factorial ratios in the classification of Bober, following the work of Vasyunin and the hypergeometric approach of Rodriguez-Villegas and Beukers–Heckman [1, 2, 4]. Our arguments require no classification of hypergeometric monodromy.

Bala’s August 2025 comments on A211417 ask for a fixed multiplier, depending on k and r but not on n , that clears the denominators in

$$\frac{A(n)}{\prod_{\substack{1 \leq i \leq r \\ (i,k)=1}} (kn+i)}, \quad k \in \{2, 3, 5\}. \quad (2)$$

There is also the negative-offset family

$$\frac{A(n)}{\prod_{\substack{1 \leq i \leq r \\ (i,30)=1}} (30n-i)}. \quad (3)$$

The current entry still labels these all- r assertions conjectural [1]. Its one-factor case $A(n)/(30n-1)$ is different: the entry records a proof in June 2026, and Adamczewski’s August 2026 report also lists it as proved [1, 5]. We use it only as a previously known special case recovered by a broader argument.

The companion A295431 entry poses similar uniform-product questions for $B(n)$, using positive slopes 1, 2, 3 and negative factors $12n-i$ with $(i, 12) = 1$ [2]. The selected problem is therefore not merely to reprove integrality of (1), or to reproduce the solved first negative-offset case. It is to explain and prove the uniform-product phenomenon.

1.2 Results obtained

Write

$$\mathcal{L}_s = \text{lcm}(1, 2, \dots, s), \quad \mathcal{L}_1 = 1, \quad m_k(r) = \#\{1 \leq i \leq r : (i, k) = 1\}.$$

Our first concrete conclusions are

$$\mathcal{L}_{30r/k}^{m_k(r)} \frac{A(n)}{\prod_{1 \leq i \leq r, (i,k)=1} (kn+i)} \in \mathbb{Z} \quad (k = 2, 3, 5), \quad (4)$$

$$\mathcal{L}_r^{m_{30}(r)} \frac{A(n)}{\prod_{1 \leq i \leq r, (i,30)=1} (30n-i)} \in \mathbb{Z}. \quad (5)$$

These hold for every $r \geq 1$ and $n \geq 0$. The indices $30r/k$ are integers. The displayed constants are deliberately explicit rather than claimed to be smallest. The corresponding negative-offset formula for B uses $\mathcal{L}_r^{m_{12}(r)}$.

The structural result is stronger. A polynomial splitting over $\mathbb{Q}$ can divide $A(n)$ uniformly up to one fixed integer multiplier precisely when it has distinct, nonzero rational roots, and the reduced denominator of each negative root lies in $\{1, 2, 3, 5\}$, while the reduced denominator of each positive root is 30. For B , these sets become $\{1, 2, 3\}$ and $\{12\}$. This also proves that the positive-offset coprimality restrictions can be removed, at the cost of changing the constant.

The eight particular constants in the two OEIS entries are correct and optimal:

Ratio	Dividing polynomial	Least positive multiplier
A	$2n + 1$	7
A	$3n + 1$	1
A	$5n + 1$	1
A	$(2n + 1)(3n + 1)(5n + 1)$	42
B	$n + 1$	385
B	$2n + 1$	5
B	$3n + 1$	1
B	$(n + 1)(2n + 1)(3n + 1)$	770

Existence for these eight cases follows already from the structural proof. Their exact constants are established by the finite certificates in Section 6.

1.3 What is and is not asserted about novelty

The source check was made on September 19, 2026. It distinguishes the still-conjectural all- r statements from the explicitly solved $30n - 1$ case. The machinery of factorial valuations and Landau step functions is classical. We give a complete proof of the additional polynomial-divisibility statements here, but do not claim that an exhaustive literature search has ruled out every equivalent formulation of the general criterion. In particular, this manuscript's mathematical claims are separable from a claim of publication priority.

2 Factorial valuations and the Landau function

2.1 Definitions and a self-contained integrality criterion

Let $a_1, \dots, a_u$ and $b_1, \dots, b_v$ be positive integers satisfying

$$\sum_{i=1}^u a_i = \sum_{j=1}^v b_j. \quad (6)$$

Set

$$R(n) = \frac{\prod_{i=1}^u (a_i n)!}{\prod_{j=1}^v (b_j n)!}, \quad \Delta(x) = \sum_{i=1}^u [a_i x] - \sum_{j=1}^v [b_j x], \quad h = v - u. \quad (7)$$

The integer h is the height. The balance condition makes Δ periodic with period one, since the terms linear in an integer translation cancel. It is a right-continuous integer-valued step function. Put $M = \max(a_i, b_j)$.

For a prime p , $v_p(q)$ denotes its exponent in a nonzero rational number q ; it may be negative. Counting the multiples of $p, p^2, \dots$ in a factorial gives

$$v_p(N!) = \sum_{e \geq 1} \lfloor N/p^e \rfloor, \quad v_p(R(n)) = \sum_{e \geq 1} \Delta(n/p^e). \quad (8)$$

All these sums are finite. A nonzero rational number is an integer if and only if all its prime valuations are nonnegative.

Lemma 2.1 (One-variable Landau criterion). *Under (6), $R(n) \in \mathbb{Z}$ for every $n \geq 0$ if and only if $\Delta(x) \geq 0$ for every real x .*

Proof. Nonnegativity gives integrality immediately from (8). Conversely, suppose Δ is negative somewhere. Right continuity and the finite set of discontinuities in a period give a nonempty open interval $(\alpha, \beta) \subset (0, 1)$ on which it is negative. Choose an arbitrarily large prime p and an integer n with $\alpha < n/p < \beta$; such an n exists once $p(\beta - \alpha) > 1$. For $p > M$ we have $0 \leq n/p^e < 1/M$ for $e \geq 2$. On this last interval every floor in (7) is zero. Thus $v_p(R(n)) = \Delta(n/p) < 0$, contradicting integrality. Periodicity handles every real x . $\square$

This is the relevant specialization of Landau's theorem; Bober gives its general setting and consequences [4, Section 3]. Bala's note supplies a direct proof for A and its floor-factorial extension [3].

2.2 Reflection and jumps

For $k \geq 1$, define

$$J(k) = \#\{i : k \mid a_i\} - \#\{j : k \mid b_j\}. \quad (9)$$

If $(t, k) = 1$, the jump at t/k is

$$\Delta(t/k) - \Delta((t/k)^-) = J(k). \quad (10)$$

Indeed, $\lfloor cx \rfloor$ increases at t/k precisely when $k \mid c$. Here $\Delta(x^-)$ is the value just to the left of x ; the value just to the right is $\Delta(x)$ itself.

Away from the discontinuities,

$$\Delta(x) + \Delta(1 - x) = h. \quad (11)$$

To see this, use $\lfloor cx \rfloor + \lfloor c(1 - x) \rfloor = c - 1$ when cx is not an integer and sum over the parameter lists. At a reduced rational point the endpoint version is

$$\Delta(t/k) + \Delta(1 - t/k) = h + J(k), \quad (12)$$

$$\Delta((t/k)^-) + \Delta((1 - t/k)^-) = h - J(k). \quad (13)$$

These follow either by counting the integral cx separately or by subtracting the two jumps. The formulas remain valid at integers by periodicity.

Corollary 2.2. *For an integral balanced ratio of height one, every value and one-sided value of Δ belongs to $\{0, 1\}$, and $J(k) \in \{-1, 0, 1\}$.*

Proof. At a point avoiding the discontinuities, the two nonnegative integers in (11) have sum one. Right continuity gives the same range at every point, and left limits have the same range. Equation (10) then bounds the jumps. $\square$

3 A complete criterion for polynomials with rational roots

3.1 The one-sided capacity

Every nonzero polynomial in $\mathbb{Z}[X]$ that splits over $\mathbb{Q}$ can be written

$$P(X) = c \prod_{j=1}^s (k_j X + \ell_j)^{m_j}, \quad (14)$$

where $c \in \mathbb{Z} \setminus \{0\}$, $k_j > 0$, $(k_j, \ell_j) = 1$, $m_j \geq 1$, and the roots $-\ell_j/k_j$ are pairwise distinct. This is primitive factorization over $\mathbb{Z}$ after collecting equal roots. Write $d = \sum_j m_j$. A constant polynomial is allowed by taking $s = 0$.

For $\ell \neq 0$, define the capacity

$$\kappa_R(k, \ell) = \begin{cases} \min_{\substack{1 \leq t \leq k \\ (t, k) = 1}} \Delta((t/k)^-), & \ell > 0, \\ \min_{\substack{1 \leq t \leq k \\ (t, k) = 1}} \Delta(t/k), & \ell < 0. \end{cases} \quad (15)$$

For $k = 1$, the only representative is $t = 1$; thus the left capacity is h and the right capacity is zero. We also set the capacity of the primitive zero-root factor X equal to zero.

The sign matters. A positive offset $kn + \ell$ forces n/p^e to approach a reduced rational point from the left. A negative offset forces approach from the right. The size of $|\ell|$ affects the exceptional small prime powers, but not the capacity.

Theorem 3.1 (Polynomial divisibility criterion). *Assume R is balanced and integral. For P as in (14), the following are equivalent:*

- (i) *There is a positive integer C such that $CR(n)/P(n) \in \mathbb{Z}$ for every $n \geq 0$ with $P(n) \neq 0$.*
- (ii) *The polynomial has no zero root and $m_j \leq \kappa_R(k_j, \ell_j)$ for every j .*

When (ii) holds and $s > 0$, one valid choice is

$$C = |c| \mathcal{L}_T^d, \quad T = \max \left(1, M \max_j |\ell_j|, \max_{i < j} |k_i \ell_j - k_j \ell_i| \right), \quad (16)$$

with the last maximum omitted if $s = 1$. For $s = 0$, $C = |c|$ works.

3.2 Proof of sufficiency

We isolate the large-prime-power part of the proof.

Lemma 3.2. *Suppose $\ell \neq 0$, $(k, \ell) = 1$, and $q = p^e > M|\ell|$. If $q \mid kn + \ell$ and $kn + \ell \neq 0$, then*

$$\Delta(n/q) \geq \kappa_R(k, \ell).$$

Proof. If $p \mid k$, then $p \nmid \ell$, so divisibility is impossible. Otherwise put $kn + \ell = tq$. Since $q > |\ell|$ and $n \geq 0$, a nonzero multiple $tq = kn + \ell$ cannot be negative; hence $t \geq 1$. Reducing modulo k gives $(t, k) = 1$, since both q and ℓ are coprime to k . We have

$$\frac{n}{q} = \frac{t}{k} - \frac{\ell}{kq}. \quad (17)$$

Every discontinuity of Δ has the form v/a_i or v/b_j . A discontinuity different from t/k is at distance at least $1/(kM)$: a nonzero difference of these rational numbers has numerator at least one after placing it over denominator ka_i or kb_j . The displacement in (17) is strictly smaller than this. Thus $\Delta(n/q)$ equals the left value at t/k if $\ell > 0$, and the right value if $\ell < 0$. Periodicity reduces t to the representatives in (15). $\square$

Sufficiency in Theorem 3.1. Let $Q(X) = \prod_j (k_j X + \ell_j)^{m_j}$. For a prime power q , put

$$E_q(n) = \sum_{j=1}^s m_j \mathbf{1}_{q \mid k_j n + \ell_j}.$$

If $q > T$, it cannot divide two distinct primitive factors. Otherwise it would divide the nonzero integer $k_i \ell_j - k_j \ell_i$, whose absolute value is at most T . Consequently, for $q > T$, either $E_q(n) = 0$ or exactly one root contributes, with its multiplicity m_j . Lemma 3.2 and (ii) give

$$\Delta(n/q) - E_q(n) \geq 0 \quad (q > T). \quad (18)$$

For $q \leq T$, Landau nonnegativity and $E_q(n) \leq d$ give the lower bound $-d$. Therefore, for every prime p ,

$$\begin{aligned} v_p(R(n)/Q(n)) &= \sum_{e \geq 1} (\Delta(n/p^e) - E_{p^e}(n)) \\ &\geq -d \# \{e \geq 1 : p^e \leq T\} = -dv_p(\mathcal{L}_T). \end{aligned}$$

This proves $\mathcal{L}_T^d R(n)/Q(n) \in \mathbb{Z}$. Since $P = cQ$, multiplying by $|c|$ only adds the harmless sign $|c|/c$. No unproved assertion about an infinite computational range enters the argument. $\square$

3.3 Proof of necessity

The converse explains why the allowed slopes are restrictive.

Necessity in Theorem 3.1. First suppose P has a zero root. Primitivity makes its factor X . For any sufficiently large prime p , take $n = p$. Then $\Delta(n/p) = \Delta(1) = 0$, and $\Delta(n/p^e) = 0$ for $e \geq 2$. Hence $v_p(R(p)) = 0$, while $p \mid P(p)$, except for finitely many p that could make $P(p) = 0$. No fixed multiplier can supply every such prime.

Now fix a nonzero root with primitive factor $kX + \ell$, multiplicity m , and capacity $\kappa < m$. Choose a reduced representative t/k at which the relevant one-sided value is κ . Dirichlet's theorem provides infinitely many primes in the progression

$$p \equiv \ell t^{-1} \pmod{k}; \quad (19)$$

this is an eligible class because $(t, k) = (\ell, k) = 1$. For $k = 1$, it means all primes. We use only the infinitude assertion of Dirichlet's theorem [6, Theorem 4.2].

For such a sufficiently large prime set

$$n = \frac{tp - \ell}{k}. \quad (20)$$

Then n is a nonnegative integer and $kn + \ell = tp$. Taking $p > k$ ensures $p \nmid t$, so the factor has valuation exactly one, and its m th power contributes m to $v_p(P(n))$. As p tends to infinity, $n/p = t/k - \ell/(kp)$ approaches from the required side; thus $\Delta(n/p) = \kappa$ eventually. For all sufficiently large p , $Mn < p^2$, so every term with $e \geq 2$ vanishes. It follows that

$$v_p(R(n)) = \kappa < m \leq v_p(P(n)).$$

The values (20) tend to infinity, so finitely many zeros of P do not obstruct the construction. Infinitely many different primes appear in reduced denominators of $R(n)/P(n)$, ruling out every fixed nonzero multiplier. $\square$

3.4 Consequences and computational meaning

The criterion is effective. All relevant values of Δ and all jumps can be computed with exact integer or rational arithmetic. It separates the problem into two very different tasks: deciding whether *some* multiplier exists, and finding the *least* multiplier. The first is answered by a finite root-by-root test. The second must account for compensation between valuation contributions at different powers of the same small prime.

The multiplicity condition is also essential. The fact that a sequence is divisible by one linear expression, up to a fixed constant, does not imply the same for its square. Higher-height ratios can support larger multiplicities: for example $A(n)^2$ supports $(2n+1)^2$, but not $(2n+1)^3$. This follows either from the theorem or from doubling the Landau function.

4 The height-one simplification

Theorem 4.1 (Jump-sign classification). *Suppose R is balanced, integral, and of height one. For a primitive factor $kX + \ell$ with $\ell \neq 0$, its capacity is one exactly when*

$$J(k) = -1 \quad \text{if } \ell > 0, \quad J(k) = +1 \quad \text{if } \ell < 0. \quad (21)$$

Otherwise its capacity is zero. Thus the admissible polynomials are exactly those with distinct nonzero roots whose primitive factors satisfy (21).

Proof. By Corollary 2.2, every value is zero or one. If $J(k) = -1$, then (10) forces the left value to be one and the right value to be zero at *every* reduced t/k . Conversely, if all left values are one, pair t/k with $1 - t/k$ in (13). Their sum is two, so $1 - J(k) = 2$ and $J(k) = -1$. The same argument with (12) proves that all right values are one exactly when $J(k) = +1$. This includes the self-paired points and $k = 1$, so no endpoint exception is needed. All capacities are at most one, and Theorem 3.1 finishes the proof. $\square$

4.1 Cyclotomic interpretation

Let $\Phi_k(z)$ be the k th cyclotomic polynomial. Factorization of $z^m - 1$ gives the rational-function identity

$$\frac{\prod_i (z^{a_i} - 1)}{\prod_j (z^{b_j} - 1)} = \prod_{k \geq 1} \Phi_k(z)^{J(k)}. \quad (22)$$

Only finitely many exponents are nonzero. At height one the allowed positive-offset denominators are exactly the cyclotomic factors remaining downstairs after cancellation in (22); the allowed negative-offset denominators are those remaining upstairs. This is an interpretation of the proved criterion, not an additional monodromy assumption.

For the binomial example $R(n) = \binom{2n}{n}$, the sets are $\{1\}$ and $\{2\}$. Accordingly, fixed products of distinct $n + i$ can be divided out up to constants, as can fixed products of distinct $2n - i$ with i odd, but a repeated such root is impossible. For the 52 sporadic ratios, Appendix A computes the two sets explicitly from their parameter lists.

5 Resolution of the uniform-product conjectures

5.1 The two exact slope lists

For A , direct cancellation in (22) gives

$$\frac{(z^{30} - 1)(z - 1)}{(z^{15} - 1)(z^{10} - 1)(z^6 - 1)} = \frac{\Phi_{30}(z)}{\Phi_1(z)\Phi_2(z)\Phi_3(z)\Phi_5(z)}. \quad (23)$$

For B ,

$$\frac{(z^{12} - 1)(z - 1)}{(z^6 - 1)(z^4 - 1)(z^3 - 1)} = \frac{\Phi_{12}(z)}{\Phi_1(z)\Phi_2(z)\Phi_3(z)}. \quad (24)$$

These identities prove the slope lists announced in Section 1, provided integrality is known. For completeness, it can be checked particularly simply here. On the grid interval $j/30 \leq x < (j+1)/30$,

$$\Delta_A(x) = j - \lfloor j/2 \rfloor - \lfloor j/3 \rfloor - \lfloor j/5 \rfloor, \quad 0 \leq j < 30.$$

The three subtracted floors have sum at most $\lfloor 31j/30 \rfloor = j$. This proves nonnegativity and hence integrality of A . For B the corresponding formula is

$$\Delta_B(x) = j - \lfloor j/2 \rfloor - \lfloor j/3 \rfloor - \lfloor j/4 \rfloor, \quad 0 \leq j < 12,$$

and the sum is at most $\lfloor 13j/12 \rfloor = j$. Height one then gives the upper bound one in each case.

Corollary 5.1. *Let $P \in \mathbb{Z}[X] \setminus \{0\}$ split over $\mathbb{Q}$. There exists a fixed positive integer C with $CA(n)/P(n) \in \mathbb{Z}$ wherever defined on $\mathbb{Z}_{\geq 0}$ if and only if P has distinct nonzero roots, every negative root in lowest terms has denominator 1, 2, 3, or 5, and every positive root in lowest terms has denominator 30. For B , replace these lists by 1, 2, 3 and 12.*

5.2 Sharper explicit bounds for the prescribed products

The general bound (16) already establishes existence. The grid structure gives the cleaner formulas (4)–(5).

Proposition 5.2. *For $k \in \{2, 3, 5\}$, let $I = \{1 \leq i \leq r : (i, k) = 1\}$ and $m = |I|$. Then the multiplier $\mathcal{L}_{30r/k}^m$ works for $A(n)/\prod_{i \in I}(kn + i)$.*

Proof. If $q = p^e > 30r/k$ divides $kn + i$, then $p \nmid k$ and $n/q = t/k - i/(kq)$ with $(t, k) = 1$. The left grid interval of width $1/30$ at t/k has $\Delta_A = 1$, since $J_A(k) = -1$. The displacement is smaller than $1/30$, so $\Delta_A(n/q) = 1$. Moreover, $q > r$ cannot divide two factors, since their difference is a nonzero integer $i - j$ of absolute value less than r . For smaller q the total deficit is at worst m . The same valuation sum used in Theorem 3.1 gives the proposed multiplier. $\square$

Proposition 5.3. *Let (R, M) be either $(A, 30)$ or $(B, 12)$ and let $I = \{1 \leq i \leq r : (i, M) = 1\}$. Then*

$$\mathcal{L}_r^{|I|} \frac{R(n)}{\prod_{i \in I}(Mn - i)} \in \mathbb{Z} \quad (n \geq 0). \quad (25)$$

Proof. No denominator factor is zero, since i is coprime to $M > 1$. If $q > r$ divides $Mn - i$, then $p \nmid M$ and $n/q = t/M + i/(Mq)$ with $(t, M) = 1$. The right grid interval of width $1/M$ at t/M has value one, because $J_R(M) = +1$. Hence the corresponding Landau term pays for this divisor. Two factors cannot both be divisible by $q > r$, because their difference has absolute value less than r . The low prime powers contribute a deficit of at most $|I|$ each, exactly as before. $\square$

For $r = 1$ the multiplier is one. Thus the proposition recovers the previously proved $A(n)/(30n - 1)$ theorem without depending on its proof. The purpose of the proposition is its uniform extension to every r . Negative values at $n = 0$ cause no problem: divisibility and valuations use the absolute value of a nonzero denominator.

5.3 Removing a restriction on positive offsets

For $R = A$ and $k \in \{1, 2, 3, 5\}$, or for $R = B$ and $k \in \{1, 2, 3\}$, consider the unrestricted product

$$P_{k,r}(n) = \prod_{i=1}^r (kn + i).$$

Put $g_i = (k, i)$ and reduce its factors to $(k/g_i)n + i/g_i$. Their roots are still distinct, and their reduced slopes divide k . Every such divisor is in the corresponding positive-offset list. Consequently a uniform multiplier exists. A fully explicit choice, with $M = 30$ or 12 respectively, is

$$\left(\prod_{i=1}^r g_i \right) \mathcal{L}_{Mr}^r. \quad (26)$$

Indeed, the primitive offsets have absolute value at most r , and their pairwise resultants are $k(j - i)/(g_i g_j)$, of absolute value less than $kr \leq Mr$. Theorem 3.1 applies, with the product of the g_i accounting for the content of the polynomial.

In contrast, the negative-offset coprimality restrictions are structural. Reducing $30n - i$ when $(30, i) > 1$ gives a proper divisor of 30 as its slope, none of which is in the negative-offset list for A . A uniform multiplier for that one factor is impossible. The same statement holds for $12n - i$ and B .

5.4 An explicit infinite obstruction family

The slope 4 is not allowed for positive offsets of A . This failure can be exhibited without relying on a table of numerical experiments. Let $p > 30$ be a prime with $p \equiv 3 \pmod{4}$, and set

$$n = \frac{3p-1}{4}.$$

Then $4n+1 = 3p$, whereas $n/p = 3/4 - 1/(4p)$ lies in $[22/30, 23/30)$, where $\Delta_A = 0$. Also $30n < p^2$ for $p > 30$, so all higher valuation terms vanish. Thus $v_p(A(n)) = 0$ while $p \mid 4n+1$. Infinitely many distinct such primes exist, so no constant clears all denominators of $A(n)/(4n+1)$. The first example in this family is $p = 31$, $n = 23$.

6 Exact optimal multipliers by finite certificates

6.1 Why checking a large interval is not enough

The large-prime-power proof gives a constant, but not necessarily the smallest one. It is not valid to maximize the deficit at each exponent independently and claim that the resulting sum is sharp: residues at $p, p^2, p^3, \dots$ are coupled by the base- p digits of n . For instance, an unfavorable contribution at p can force a favorable contribution at p^2 or later. This is why some quotients above need no multiplier even though an individual Landau summand can fail to cover the denominator at a small prime power.

We encode all possible digit continuations in a finite directed graph. Its integer edge weights add to the exact valuation, not to a floating point approximation. A feasible potential on the graph proves a lower bound for every possible n simultaneously.

6.2 The state and transition rules

For this section let $R = A$ or B , let $L = 30$ or 12 respectively, and let

$$Q(n) = \prod_{k \in K} (kn + 1),$$

where K is one of the eight slope sets in the result table. Write $\delta(j) = \Delta_R(j/L)$ for $0 \leq j < L$. Process a base- p expansion of n from least significant digit to most significant digit. After e digits, let $r_e = n \bmod p^e$ and set

$$j_e = \lfloor Lr_e/p^e \rfloor.$$

Because every factorial parameter divides L , the valuation summand $\Delta_R(n/p^e)$ equals $\delta(j_e)$.

A factor $kn + 1$ is called alive after e digits when $p^e \mid kr_e + 1$. For an alive factor retain the integer carry

$$t_{k,e} = (kr_e + 1)/p^e.$$

Its possible values are $1, \dots, k$. Once a factor is not alive, it can never be alive again, since divisibility by p^{e+1} implies divisibility by p^e . Mark it by $\perp$ and forget its carry. The initial state is

$$s_0 = (0, 1, \dots, 1).$$

If the next digit is $d \in \{0, \dots, p-1\}$, the next state satisfies

$$j' = \lfloor (j + Ld)/p \rfloor, \quad (27)$$

and for each factor

$$t'_k = \begin{cases} (t_k + kd)/p, & t_k \neq \perp \text{ and } p \mid t_k + kd, \\ \perp, & \text{otherwise.} \end{cases} \quad (28)$$

To verify (27), write $Lr_e/p^e = j + \varepsilon$ with $0 \leq \varepsilon < 1$; dividing $j + Ld + \varepsilon$ by the integer p does not alter the floor obtained from $j + Ld$. The carry formula follows by replacing r_e with $r_e + dp^e$.

Give this transition the weight

$$w(s, d) = \delta(j') - \#\{k \in K : t'_k \neq \perp\}. \quad (29)$$

Reading all the digits of n and then enough zero digits reaches

$$s_* = (0, \perp, \dots, \perp).$$

Indeed $j_e = 0$ when $p^e > Ln$, and no positive integer $kn + 1$ is divisible by every power of p . The terminal state remains fixed under further zero digits and has zero edge weight. Equations (8) and (29) show that the sum along this finite path is exactly

$$v_p(R(n)/Q(n)). \quad (30)$$

6.3 Potential certificates

Proposition 6.1 (Finite certificate rule). *Let S be a finite collection of states containing s_0 and s_* and closed under every digit transition. Suppose an integer function $H : S \rightarrow \mathbb{Z}$ satisfies*

$$H(s_*) = 0, \quad H(s) \leq w(s, d) + H(s') \quad \text{for every } s \in S \text{ and every digit } d. \quad (31)$$

Then $v_p(R(n)/Q(n)) \geq H(s_0)$ for every $n \geq 0$. If some explicit n_0 has valuation $H(s_0)$, the lower bound is sharp.

Proof. Apply the edge inequality successively along the path for n described above. The intermediate values of H telescope, leaving $H(s_0) \leq \sum w + H(s_*) = v_p(R(n)/Q(n))$. An equality witness proves that no larger lower bound is possible. $\square$

The generator enumerates reachable states and computes shortest costs to the terminal by integer relaxation. This search is *not* trusted as the proof: a separate program reads the resulting state list and potential and checks every closure condition and inequality in (31). It also checks the equality witness directly from Legendre's formula. All prime certificates needed for a particular quotient are checked for complete coverage.

6.4 Results of the exact verification

For A the general cutoff (16) is $T = 30$ for these positive unit offsets; for B it is $T = 12$. Thus primes larger than these bounds need no certificate: each of their valuation summands is nonnegative after subtracting the denominator indicator. Below the cutoff the exact minima are as follows. A zero beyond a row's cutoff is both a lower bound from the theorem and an attained minimum at $n = 0$.

$(R; K)$	2	3	5	7	11	13	17	19	23	29	$C_{\min}$
$(A; 2)$	0	0	0	-1	0	0	0	0	0	0	7
$(A; 3)$	0	0	0	0	0	0	0	0	0	0	1
$(A; 5)$	0	0	0	0	0	0	0	0	0	0	1
$(A; 2, 3, 5)$	-1	-1	0	-1	0	0	0	0	0	0	42
$(B; 1)$	0	0	-1	-1	-1	0	0	0	0	0	385
$(B; 2)$	0	0	-1	0	0	0	0	0	0	0	5
$(B; 3)$	0	0	0	0	0	0	0	0	0	0	1
$(B; 1, 2, 3)$	-1	0	-1	-1	-1	0	0	0	0	0	770

The headings $2, 3, \dots$ are primes, and the entries are $\min_{n \geq 0} v_p(R(n)/Q(n))$. The table is not inferred from testing an interval: it is the output of the checked finite potentials.

The nonzero lower-bound requirements are all necessary, as witnessed by the following small indices:

$(R; K)$	Prime	Index with valuation -1
$(A; 2)$	7	10
$(A; 2, 3, 5)$	2, 3, 7	3, 16, 10, respectively
$(B; 1)$	5, 7, 11	9, 20, 10, respectively
$(B; 2)$	5	2
$(B; 1, 2, 3)$	2, 5, 7, 11	1, 2, 20, 10, respectively

For each prime, any valid multiplier must contain that prime to at least the indicated deficit. Multiplying these independently necessary prime powers gives the least multiplier; the indices need not be the same for different primes. This proves optimality of all eight constants.

The delivered certificates comprise **60 prime graphs, 1,685 states, and 19,277 checked transition inequalities**. Their compact JSON representation contains every state and every potential value, together with equality witnesses. The separate verifier recomputes all transitions instead of trusting stored edges.

7 Integer sequences, recurrences, and generating functions

7.1 An integer sequence singled out by the optimal constant

Define

$$U_n = \frac{42A(n)}{(2n+1)(3n+1)(5n+1)}. \quad (32)$$

The preceding section proves $U_n \in \mathbb{Z}_{>0}$, with the factor 42 minimal for integrality at every $n \geq 0$. Its first terms are

$$\begin{aligned} U_0 &= 42, \\ U_1 &= 45287852610, \\ U_2 &= 5873158887743776788432, \\ U_3 &= 1645528285131615053352729526247883. \end{aligned}$$

The archive supplies exact terms through $n = 50$. We make no assertion that this sequence lacks an existing OEIS entry; the contribution here is its proved uniform integrality and optimal normalization.

For an arbitrary polynomial divisor P covered by our theorem, $u_n = CR(n)/P(n)$ has the explicit first-order recurrence

$$\frac{u_{n+1}}{u_n} = \frac{P(n)}{P(n+1)} \frac{\prod_{i=1}^u \prod_{t=1}^{a_i} (a_i n + t)}{\prod_{j=1}^v \prod_{t=1}^{b_j} (b_j n + t)}. \quad (33)$$

Cross-multiplication gives a recurrence with polynomial coefficients. This follows directly by canceling consecutive factorials and is valid wherever the quotient is defined. In our admissible cases P has no nonnegative integer root, so there is no missing index.

7.2 A hypergeometric representation

Set

$$\lambda = 2^{14} 3^9 5^5, \quad \alpha = \frac{1}{30}(1, 7, 11, 13, 17, 19, 23, 29),$$

and let the seven denominator parameters be

$$\beta = \left(\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5} \right).$$

Cancellation of the linear factors in $A(n+1)/A(n)$ gives

$$A(n) = \lambda^n \frac{\prod_{a \in \alpha} (a)_n}{n! \prod_{b \in \beta} (b)_n}, \quad (34)$$

where $(a)_n = a(a+1) \cdots (a+n-1)$ and $(a)_0 = 1$. One can verify the identity without invoking the gamma multiplication formula: both sides have value one at $n = 0$ and the same ratio of successive terms. It is also consistent with the hypergeometric description in A211417 [1].

The elementary identity

$$\frac{1}{kn+1} = \frac{(1/k)_n}{(1+1/k)_n}$$

shows that dividing by the three factors in (32) shifts three denominator parameters. Define

$$\beta' = \left(\frac{3}{2}, \frac{4}{3}, \frac{2}{3}, \frac{6}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5} \right).$$

Then

$$U_n = 42\lambda^n \frac{\prod_{a \in \alpha} (a)_n}{n! \prod_{b \in \beta'} (b)_n}, \quad (35)$$

and hence

$$G(z) := \sum_{n \geq 0} U_n z^n = 42 {}_8F_7 \left(\begin{matrix} \frac{1}{30}, \frac{7}{30}, \frac{11}{30}, \frac{13}{30}, \frac{17}{30}, \frac{19}{30}, \frac{23}{30}, \frac{29}{30} \\ \frac{2}{3}, \frac{3}{3}, \frac{5}{5}, \frac{6}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5} \end{matrix}; \lambda z \right). \quad (36)$$

This is an equality of formal power series, and of analytic functions for $|\lambda z| < 1$. The coefficients also satisfy the convenient recurrence

$$(n+1) \prod_{b \in \beta'} (n+b) U_{n+1} = \lambda \prod_{a \in \alpha} (n+a) U_n, \quad U_0 = 42. \quad (37)$$

No recurrence is being used to define an unproved integer sequence: integrality has already been established independently.

7.3 Differential and integral transforms

Let $\theta = z \, d/dz$ and $F(z) = \sum_{n \geq 0} A(n)z^n$. Coefficient extraction gives

$$(2\theta + 1)(3\theta + 1)(5\theta + 1)G(z) = 42F(z). \quad (38)$$

Equivalently, within the common disk of convergence,

$$G(z) = 42 \int_0^1 \int_0^1 \int_0^1 F(zx^2y^3t^5) \, dx \, dy \, dt. \quad (39)$$

Termwise integration is justified by absolute convergence, and each monomial contributes the three reciprocal factors in (32).

The first-order recurrence (37) gives the differential equation

$$\left[\theta \prod_{b \in \beta'} (\theta + b - 1) - \lambda z \prod_{a \in \alpha} (\theta + a) \right] G(z) = 0. \quad (40)$$

For $n \geq 1$, its z^n coefficient is just the recurrence with index $n - 1$; its constant coefficient vanishes due to the initial factor θ . Thus G is D-finite. More generally, polynomial division of coefficients gives $P(\theta) \sum u_n z^n = C \sum R(n) z^n$. These identities make the new integer divisibility statements useful for recurrence and generating-function investigations, without needing to guess a closed algebraic equation.

8 All-order asymptotic expansions

8.1 The general factorial-ratio expansion

The usual Stirling expansion for $\log \Gamma$, including remainder control on the positive real axis, is recorded in DLMF Section 5.11 [8]. Applying it separately to the finitely many factorials in $R(n)$ yields the following explicit form. Let

$$\Lambda = \frac{\prod_i a_i^{a_i}}{\prod_j b_j^{b_j}}, \quad K_R = (2\pi)^{-h/2} \sqrt{\frac{\prod_i a_i}{\prod_j b_j}},$$

and, with B_{2q} the Bernoulli numbers, set

$$\sigma_{2q-1} = \frac{B_{2q}}{2q(2q-1)} \left(\sum_i a_i^{1-2q} - \sum_j b_j^{1-2q} \right). \quad (41)$$

Then, for any fixed $N \geq 1$,

$$\log R(n) = n \log \Lambda - \frac{h}{2} \log n + \log K_R + \sum_{q=1}^N \frac{\sigma_{2q-1}}{n^{2q-1}} + O(n^{-2N-1}). \quad (42)$$

The terms of size $n \log n$ and n cancel by balance, the $\log n$ terms give $-h/2$, and the remaining terms are exactly those displayed. Thus (42) is a Poincare expansion to any prescribed order, not a claim that its infinite series converges.

Suppose P has degree d , positive leading coefficient ρ , and factorization (14) with all roots nonzero. For sufficiently large n , it is positive. Write

$$S_j = \sum_{i=1}^s m_i (\ell_i/k_i)^j.$$

Expanding the finitely many logarithms in $P(n) = \rho n^d \prod_i (1 + \ell_i/(k_i n))^{m_i}$ gives

$$\log P(n) = \log \rho + d \log n + \sum_{j=1}^N \frac{(-1)^{j+1} S_j}{j n^j} + O(n^{-N-1}). \quad (43)$$

Subtracting (43) from (42) proves an all-order expansion for every integer sequence constructed here. For example the first two correction terms are

$$\frac{CR(n)}{P(n)} = \frac{CK_R}{\rho} \Lambda^n n^{-d-h/2} \left[1 + \frac{\sigma_1 - S_1}{n} + \frac{(\sigma_1 - S_1)^2 + S_2}{2n^2} + O(n^{-3}) \right]. \quad (44)$$

For a negative leading coefficient, the same assertion holds with its sign restored; the logarithmic derivation is applied to $|P(n)|$.

8.2 The expansion for the certified sequence

For A one obtains

$$K_A = \frac{1}{\sqrt{60\pi}}, \quad \Lambda = \lambda, \quad \sigma_1 = \frac{7}{120}.$$

In (32), $d = 3$, $\rho = 30$,

$$S_1 = \frac{1}{2} + \frac{1}{3} + \frac{1}{5} = \frac{31}{30}, \quad S_2 = \frac{1}{4} + \frac{1}{9} + \frac{1}{25} = \frac{361}{900}.$$

Therefore

$$U_n = \frac{7}{5\sqrt{60\pi}} \frac{\lambda^n}{n^{7/2}} \left(1 - \frac{39}{40n} + \frac{3893}{5760n^2} + O(n^{-3}) \right). \quad (45)$$

All further coefficients are obtained by the explicit rational operations in (41)–(43). For B , the analogous base constants are $\Lambda = 2^{10}3^3$, $K_B = 1/\sqrt{12\pi}$, and $\sigma_1 = 1/36$. The method therefore gives comparably explicit expansions for every companion sequence in the sharp-constant table.

9 Verification record, limitations, and next questions

9.1 What was actually checked

The archive contains two different kinds of computational work. The finite certificates are part of the proof of the exact constants; the regression tests merely provide additional protection against implementation and transcription errors. The separate certificate checker completed successfully for

all 60 graphs, all 1,685 states, and all 19,277 transition inequalities, including prime coverage and sharpness witnesses.

The supplementary tests passed 90,298 comparisons between two forms of the Landau step function, 808 direct rational checks of the eight sharp quotients, and 480,048 small-prime valuation checks over $0 \leq n \leq 10,000$. They also passed 3,336 direct checks of the explicit uniform-product bounds, 4,000 valuation checks at reproducibly sampled 256-bit indices, and 668 classification checks. The random seed was 20260919. A further 203 exact checks verify the hypergeometric recurrences and the first asymptotic coefficients. These tests are recorded in `data/test_results.json`; they are not cited as substitutes for Theorem 3.1.

Every sporadic parameter list in the atlas was separately checked for balance, height one, and nonnegativity at all rational breakpoints of its Landau function. The atlas’s parameter order follows Coserea’s OEIS table [7]. The derived jump-sign columns are generated from (9), not inferred from a finite list of sequence values.

9.2 Limits of the conclusions

The polynomial criterion applies to polynomials splitting over $\mathbb{Q}$. It does not classify arbitrary irreducible quadratic or higher-degree polynomials. Nor does the height-one sign rule extend unchanged to higher height: there one must use the actual capacities in (15), and multiplicities greater than one may occur. The bound (16) is uniform and constructive, but often much larger than the minimum; optimal constants for general products remain a useful computational problem.

No Lean formalization was produced or run. The general proofs are written in conventional mathematics, and the finite-graph certificates are checked by ordinary exact-integer Python code. There is no reliance on floating-point root finding, numerical asymptotics, or a conjectured recurrence in the proof of integrality. A Lean development could split naturally into the valuation identity, rational one-sided step lemmas, the prime-power cutoff argument, and a generic finite-potential checker.

9.3 Research directions arising from the proof

A natural continuation is to classify nonsplit polynomial divisors. Our necessity proof realizes prescribed rational approach points by primes in arithmetic progressions. For higher-degree factors, roots modulo primes and their distribution take the place of these simple progressions, so the rational-root proof cannot simply be copied.

A second question is the behavior, as r grows, of the *optimal* uniform multipliers for the two product families. The explicit lcm bounds settle existence but do not predict the exact small-prime exponents. The finite-state method suggests an effective route for fixed r , with a separate challenge of controlling the state space uniformly in r . Finally, the newly certified sequences invite combinatorial models that would explain their integrality without prime valuations; no such interpretation is assumed here.

Part II. Rational dilation, integrality, and algebraicity of factorial ratios

Proofs for A347854–A347858, a second route to Part I’s A295431 multipliers, and a divisor theorem for A295432

10 Part II: source, overlap with Part I, and notation

10.1 Source and status

Part II is the text of one manuscript, *Rational Dilation, Integrality, and Algebraicity of Factorial Ratios* (“Research draft prepared with ChatGPT”, dated October 1, 2026), delivered as the archive `OEIS_Fractional_Factorial_Theorems.zip` in commit `c664fc2f0`. It is manuscript 01 of batch 74 of ProveIt’s incoming-reports intake and was placed beside Part I in commit `b669cff87`. It names no ProveIt commit as its pin; its only repository citation is the root `README.md` blob `bfe39c17`, introduced by commit `e37a17848` [20]. Every result, proof, remark, limitation and research question of the manuscript is printed below. Its abstract reads as follows, with the sentence on A295431 corrected as explained in Section 10.2 and the Lambert branch glossed:

We give ordinary mathematical proofs of the integrality and algebraicity conjectures attached to the five fractional-factorial sequences A347854–A347858. The arithmetic argument does not apply Landau’s criterion to nonintegral factorial arguments. Instead, it first cancels the Gamma quotient to an integer arithmetic progression, derives an exact residue-dependent prime-valuation formula, and treats primes dividing the fractional denominator separately. This yields a general rational-dilation theorem, valid for composite as well as prime denominators.

For algebraicity, the fractional subsequences are identified with Frobenius solutions of four finite-monodromy hypergeometric equations. Four small, positive-definite integral Gram matrices provide exact finiteness certificates. We also give a second proof of the specified denominator-clearing conjectures posted in August 2025 in A295431: four specified multipliers, division by $12n - 1$, and two infinite families with explicit clearing constants. Part I had already proved all of them, with smaller constants and with the four multipliers shown least possible. A further product-divisibility conjecture in A295432 and the June 2026 conditional refinements of the first two fractional sequences follow.

Supplementary results include beta-product moment representations, strict Hankel positivity, all-orders asymptotic expansions with effective remainders, and inverse expansions based on the negative real branch of Lambert’s W function (the lower real branch W_{-1} , with values at most -1 , which yields the large solution). A standard-library Python program checks the finite certificates and additional exact-arithmetic identities. These checks supplement, rather than replace, the proofs.

Status. Unrefereed research manuscript. No new theorem in this package has been checked in Lean or Rocq, and no OEIS submission has been made. The classical monodromy results used are identified explicitly. The search supports the reported OEIS conjecture status, not a claim of worldwide priority.

10.2 What Part II re-proves from Part I

The manuscript was written without reference to Part I and does not cite it. Its statements on A295431 are therefore not new. Theorem 11.2 (A295431 part) re-proves results of Part I:

- The multipliers 385, 5, 1, 770 for $n + 1$, $2n + 1$, $3n + 1$ and their product are proved in Part I and shown least possible there (the table of least multipliers in Section 6 and its finite digit-graph certificates). Part I's $B(n)$ is the sequence called $U(n)$ here.
- The negative family is Part I's Proposition 5.3 with $(R, M) = (B, 12)$, whose constant $\mathcal{L}_r^{|I|}$ divides the constant $D(r) = L_r^r$ used here; its case $r = 1$ is the divisibility of $U(n)$ by $12n - 1$ with multiplier one.
- The positive family is Part I's unrestricted-product bound (26) with $M = 12$, whose constant $(\prod_{i \leq r} g_i) \mathcal{L}_{12r}^r$ divides the constant $C(k, r) = (kL_{12r})^r$ used here, since every $g_i = (k, i)$ divides k .

The digit-sum proofs of Sections 16.2 and 16.3 are a second, independent route to these statements, and are printed as such. The proof of $U(n)/(12n - 1) \in \mathbb{Z}$ through Theorem 15.1 is a third route to that one case.

10.3 What is new in Part II

Relative to Part I and to the rest of the repository at the time of writing, the following are new:

- integrality and algebraicity of the generating functions of A347854–A347858 (Theorem 11.1); Part I's status notes had left the Gamma-interpolated values of this OEIS family explicitly unsettled;
- the rational-dilation theorem with composite denominators (Theorem 12.1), the residue-shifted valuation formula, its reduction to slices of the ordinary Landau function, and the treatment of the denominator primes (Sections 13 and 14);
- the June 2026 refinements for A347854 and A347855 (Section 15);
- the divisor $6(6n + 1)(12n - 1)$ of A295432, (54), proved in Section 16.4;
- the Gram-matrix finite-monodromy certificates, the beta-product moment laws and strict Hankel positivity (Sections 18 and 19);
- the explicit fixed-order remainder constant E_J in Theorem 20.1.

The rest of Sections 20 and 21 instantiates expansions that are already in the repository: Part I's factorial-ratio expansion (42), and the gamma-quotient engine and balanced inverse theorem of the transseries volumes [21, 22]; see the notes in those sections. Part II claims no novelty there.

10.4 Notation

Each Part keeps its own definitions. Several letters mean different things in the two Parts, and some mean two things within Part II, as in the source. The following table fixes the reading. Within Part II every symbol has its Part II meaning.

Symbol	Part I	Part II
$B(n)$	A295431, $(12n)!n!/((6n)!(4n)!(3n)!)$	not used for a sequence; A295432 is $\tilde{B}(n)$, and A295431 is $U(n)$
$\tilde{B}(n)$	—	A295432, $(12n)!(3n)!(2n)!/((6n)!^2(4n)!n!)$; the source writes $B(n)$
U	$U_n = 42A(n)/((2n+1)(3n+1)(5n+1))$	$U(n) = A295431 = \text{Part I's } B(n)$; U_d the general ordinary ratio; $U_I, \dots, U_{IV}$ the four bases
A, B	A211417 and A295431	integer slope parameters in $L = A + B + m$; $A_i(x)$ generating functions
$\mathcal{L}_s, L_r$	$\mathcal{L}_s = \text{lcm}(1, \dots, s)$	$L_r = \text{lcm}(1, \dots, r)$ (the same function); L alone is a slope parameter
valuation	v_p	ν_p (same meaning)
Δ	Landau function of R	Landau function of U_d , Δ_U , $\Delta_{\tilde{B}}$; Δ^k finite differences in Theorem 19.1
R	a generic factorial ratio	exponential growth constant R (and R_i)
λ, K	$\lambda = 2^{14}3^95^5$; K_R	λ per base; K amplitude constant
h	height	length of the reduced parameter lists; $\nu_p(n)$ in Section 14.2
$m, m_k(r)$	$m_k(r)$ counts units	m the integer gap in $L = A + B + m$
P, Q	divisor polynomial; $Q = P/c$	$P(T), Q(T)$ of the operator (80)
T	cutoff in (16)	polynomial variable; T_n a multinomial; T transpose
J	jump $J(k)$	expansion order in Theorem 20.1; base index
σ, γ	σ_{2q-1}	γ_{2j-1} plays the role of Part I's σ_{2q-1}
B, B_{2j}	B_{2q} Bernoulli	B_{2j} Bernoulli; $B(\cdot, \cdot)$ the beta function

The tempting false reading is that Part II's B or U is Part I's: it is not. Part I's B is Part II's U , and Part I's U_n does not occur in Part II. The only renaming made in printing the source is $B(n) \mapsto \tilde{B}(n)$ for A295432; the valuation symbol is printed ν_p throughout, as in the source.

10.5 Reading guide

The arithmetic proof is self-contained in Sections 12 to 16. Section 13 derives the residue formula from a finite rational product; Section 14 treats all primes, including those dividing a composite denominator. Section 16 proves the specified multipliers and the two infinite denominator-clearing families (for A295431 a second route to Part I; see Section 10.2).

The algebraicity proof has a separate dependency: the classical companion-matrix description of hypergeometric monodromy. Section 17 explains the fractional Frobenius solutions, and Section 18 gives the four exact matrix certificates. Sections 19 to 21 develop moment laws, forward asymptotics, and inverse expansions. The verification discussion in Section 22 separates finite proof certificates from finite numerical tests.

Appendix C maps each OEIS target to its proof. The shipped files are listed in Appendix D.

11 The questions and the contribution

11.1 Five related sequences

Throughout, $x!$ for nonintegral $x \geq 0$ means $\Gamma(x+1)$. Define

$$a_1(n) = \frac{(6n)!(n/2)!}{(3n)!(2n)!(3n/2)!}, \quad \text{A347854,} \quad (46)$$

$$a_2(n) = \frac{(4n)!(n/3)!}{(2n)!n!(4n/3)!}, \quad \text{A347855,} \quad (47)$$

$$a_3(n) = \frac{(6n)!(n/2)!}{(4n)!n!(3n/2)!}, \quad \text{A347856,} \quad (48)$$

$$a_4(n) = \frac{(6n)!(3n/2)!}{(3n)!(2n)!(5n/2)!}, \quad \text{A347857,} \quad (49)$$

$$a_5(n) = \frac{(9n)!(n/2)!}{(3n)!(2n)!(9n/2)!}, \quad \text{A347858.} \quad (50)$$

Here and below n is a nonnegative integer unless another domain is stated. The inspected OEIS entries conjecture that the values are integers and that the ordinary generating functions are algebraic [9, 10, 11, 12, 13].

Theorem 11.1 (Five integrality and five algebraicity assertions). *For $1 \leq i \leq 5$ and $n \geq 0$, $a_i(n)$ is a positive integer. Each series*

$$A_i(x) = \sum_{n \geq 0} a_i(n)x^n$$

is algebraic over $\mathbb{Q}(x)$.

These examples are not isolated numerical curiosities. They are rational samplings of four of the sporadic height-one factorial ratios related to finite hypergeometric monodromy. The ordinary ratios and their place in that family are recorded in A295431 and the cited literature [14, 4, 16]. Our contribution is the arithmetic and differential bridge from ordinary integer arguments to the fractional arguments, together with concrete denominator-clearing results.

Sequence	$a_i(1)$	$a_i(2)$	$a_i(3)$	$a_i(4)$
A347854	40	4620	622336	89237148
A347855	9	189	4620	120285
A347856	20	990	56576	3432198
A347857	24	1386	91392	6374082
A347858	512	1021020	2399141888	6016814703900

The table fixes the normalization; it is not evidence sufficient for the theorem.

11.2 Further conjectures: A295431 (a second route to Part I) and A295432

Put

$$U(n) = \frac{(12n)!n!}{(6n)!(4n)!(3n)!}, \quad \tilde{B}(n) = \frac{(12n)!(3n)!(2n)!}{(6n)!^2(4n)!n!}. \quad (51)$$

These are A295431 and A295432, respectively. The following statements match the relevant denominator and divisibility conjectures in those entries [14, 15]. The sequence U is Part I's B ; the source writes $B(n)$ for $\tilde{B}(n)$ (Section 10.4).

Theorem 11.2 (Specified clearing factors and two infinite families; the A295431 part is Part I's). *For every $n \geq 0$, all five quantities*

$$\frac{385U(n)}{n+1}, \quad \frac{5U(n)}{2n+1}, \quad \frac{U(n)}{3n+1}, \quad \frac{770U(n)}{(n+1)(2n+1)(3n+1)}, \quad \frac{U(n)}{12n-1} \quad (52)$$

are integers. For $r \geq 1$, define

$$L_r = \text{lcm}(1, 2, \dots, r), \quad C(k, r) = (kL_{12r})^r, \quad D(r) = L_r^r.$$

Then, for $k \in \{1, 2, 3\}$ and $n \geq 0$,

$$\frac{C(k, r)U(n)}{\prod_{i=1}^r (kn+i)} \in \mathbb{Z}, \quad \frac{D(r)U(n)}{\prod_{\substack{1 \leq i \leq r \\ (i, 12)=1}} (12n-i)} \in \mathbb{Z}. \quad (53)$$

In addition, for $n \geq 1$,

$$6(6n+1)(12n-1) \mid \tilde{B}(n). \quad (54)$$

The constants in (53) are deliberately simple, not claimed minimal. The specified multipliers in (52) are the multipliers conjectured in the entry; their optimality is not needed or asserted.

[write] **Added 1 October 2026, batch 74.** Everything in (52) and (53) was already proved in Part I, which the source did not consult. Part I proves the four multipliers least possible (Section 6), proves $U(n)/(12n-1) \in \mathbb{Z}$ as the case $r = 1$ of Proposition 5.3, and gives the smaller family constants $\mathcal{L}_r^{|I|}$ (Proposition 5.3) and $(\prod_{i \leq r} g_i) \mathcal{L}_{12r}^r$ in (26), which divide $D(r)$ and $C(k, r)$ respectively. The proofs in Sections 16.2 and 16.3 are a second route (Section 10.2). Only (54), on A295432, is new.

11.3 Provenance and boundaries

The source audit used the ProveIt repository and the current OEIS entries. The repository's distinction between research artifacts, executable checks, and trusted formal theorems is adopted here [20]. No theorem from an unreviewed repository report is assumed in the arguments below. A repository search for the principal target did not return a matching file; this is not an exhaustive priority search.

[write] **Added 1 October 2026, batch 74.** The search was right about A347854–A347858 and A295432, which no repository file mentioned. It missed this report's Part I (in the repository since September 2026), which proves the A295431 statements of Theorem 11.2 with smaller constants and with optimality; see Section 10.2. The cited repository source [20] is the root README.md blob bfe39c17 of commit e37a17848. The first-party account [19] cited next is an external web page, not a repository file.

An August 2026 first-party account records a failed proof attempt for the same fractional-factorial family, based on applying an integer-parameter floor criterion outside its hypotheses [19]. The underlying obstruction is mathematical, not computational: checking a floor inequality exactly does not establish that it computes the valuations of a Gamma ratio. Section 13 provides the missing bridge and handles the exceptional primes explicitly.

The leading asymptotic scale and some moment representations already appear in the OEIS material. They are not advertised as discoveries here. The primary results are the arithmetic, algebraicity, and denominator-clearing proofs (for A295431, a second route to Part I). The supercongruences appearing in these entries are *not* proved in this manuscript. Neither explicit minimal algebraic equations nor their degrees are determined.

12 A rational-dilation theorem

Let L, A, B, m, c, d be positive integers satisfying

$$L = A + B + m, \quad (c, d) = 1. \quad (55)$$

For real $t \geq 0$, define

$$V(t) = \frac{\Gamma(1 + Lt)\Gamma(1 + ct/d)}{\Gamma(1 + At)\Gamma(1 + Bt)\Gamma(1 + (c/d + m)t)}, \quad (56)$$

$$U_d(t) = V(dt) = \frac{\Gamma(1 + dLt)\Gamma(1 + ct)}{\Gamma(1 + dAt)\Gamma(1 + dBt)\Gamma(1 + (c + dm)t)}. \quad (57)$$

The subscript distinguishes this general ordinary ratio from the particular U in (51).

Theorem 12.1 (Rational dilation, including composite denominators). *Assume (55) and $U_d(n) \in \mathbb{Z}$ for all $n \geq 0$. Then*

$$V(n) \in \mathbb{Z}_{>0}, \quad \left(\frac{d}{\gcd(d, n)} \right)^{mn} \mid V(n) \quad (n \geq 0). \quad (58)$$

Furthermore, $\sum_{n \geq 0} V(n)x^n$ is algebraic over $\mathbb{Q}(x)$.

The arithmetic proof is given in Sections 13–14. The algebraicity assertion is proved in Section 17. The special orientation of the Gamma quotient in (56), the positive integer gap mn , and the balance $L = A + B + m$ are essential to the stated proof. This is not a criterion for arbitrary rational slopes.

The five sequences correspond to the following parameters.

Sequence	L	A	B	m	c	d	ordinary base
A347854	6	3	2	1	1	2	I
A347855	4	2	1	1	1	3	I
A347856	6	4	1	1	1	2	II
A347857	6	3	2	1	3	2	III
A347858	9	3	2	4	1	2	IV

The bases are

$$\begin{aligned} U_{\text{I}}(t) &= \frac{(12t)!t!}{(6t)!(4t)!(3t)!}, \\ U_{\text{II}}(t) &= \frac{(12t)!t!}{(8t)!(2t)!(3t)!}, \\ U_{\text{III}}(t) &= \frac{(12t)!(3t)!}{(6t)!(4t)!(5t)!}, \\ U_{\text{IV}}(t) &= \frac{(18t)!t!}{(6t)!(4t)!(9t)!}. \end{aligned} \quad (59)$$

They are A295431, A295433, A295435, and A295437. The relation is $a_i(n) = U_J(n/d)$, with J, d taken from the table.

13 The exact prime-valuation bridge

13.1 First eliminate the fractional Gamma arguments

For integer $n \geq 0$, the functional equation of Gamma gives the finite identity

$$V(n) = \frac{d^{mn}(Ln)!}{(An)!(Bn)! \prod_{j=1}^{mn} (cn + dj)}. \quad (60)$$

In particular, $V(n)$ is a positive rational number before integrality is proved. Every valuation below is a valuation of this rational number. We never assign a prime valuation to an individual transcendental Gamma value.

For $s \in \{0, \dots, d-1\}$ and $0 \leq x < 1$, set

$$F_s(x) = \lfloor Lx \rfloor - \lfloor Ax \rfloor - \lfloor Bx \rfloor - \left\lfloor \frac{(c + dm)x - s}{d} \right\rfloor + \left\lfloor \frac{cx - s}{d} \right\rfloor. \quad (61)$$

The dependence on s is not optional: it records a congruence class in the arithmetic progression in (60).

Lemma 13.1 (Exact valuation formula away from the denominator). *Let p be prime with $p \nmid d$. For each $k \geq 1$, put $q = p^k$, write $n = qt + r$ with $0 \leq r < q$, and choose $s_k \in \{0, \dots, d-1\}$ such that*

$$qs_k \equiv cr \pmod{d}. \quad (62)$$

Then

$$\nu_p(V(n)) = \sum_{k \geq 1} F_{s_k} \left(\frac{n \bmod p^k}{p^k} \right). \quad (63)$$

Only finitely many summands are nonzero.

Proof. At level q , Legendre's formula gives the factorial contribution

$$\left\lfloor \frac{Ln}{q} \right\rfloor - \left\lfloor \frac{An}{q} \right\rfloor - \left\lfloor \frac{Bn}{q} \right\rfloor = mt + \left\lfloor \frac{Lr}{q} \right\rfloor - \left\lfloor \frac{Ar}{q} \right\rfloor - \left\lfloor \frac{Br}{q} \right\rfloor.$$

For the denominator progression, $q \mid cn + dj$ is equivalent to $q \mid cr + dj$. Since $(q, d) = 1$, exactly one residue class of j modulo q is counted. Among $mn = mqt + mr$ consecutive indices there are mt complete blocks of length q , followed by mr indices.

In the latter block, write $cr + dj = q\ell$. With $x = r/q$, the inequalities $1 \leq j \leq mr$ are precisely

$$cx < \ell \leq (c + dm)x.$$

Integrality of j is precisely $\ell \equiv s_k \pmod{d}$. The number of such ℓ is

$$\left\lfloor \frac{(c + dm)x - s_k}{d} \right\rfloor - \left\lfloor \frac{cx - s_k}{d} \right\rfloor.$$

Subtracting this count and the mt complete blocks from the factorial contribution gives $F_{s_k}(x)$. Summing over $q = p^k$ proves the identity. For $q > \max(L, c + dm)n$, no numerator factorial or progression factor is divisible by q , so the level contribution vanishes. $\square$

13.2 Why a naive rational-slope floor criterion fails

For $x \geq 0$ one has

$$\lfloor 7x/6 \rfloor - \lfloor x/6 \rfloor - \lfloor x \rfloor \geq 0,$$

because $7x/6 = x/6 + x$. Nevertheless,

$$\frac{\Gamma(1 + 7/6)}{\Gamma(1 + 1/6)\Gamma(2)} = \frac{7}{6} \notin \mathbb{Z}.$$

Thus balance and this unshifted floor inequality do not prove integrality of the corresponding Gamma quotient. The same counterexample is documented in the failed-attempt account [19]. Our argument starts with the rational identity (60), derives (63), and then addresses the primes excluded from that formula. Merely increasing the number of sampled terms would not repair the invalid shortcut.

14 Ordinary Landau functions and all denominator primes

14.1 The ordinary criterion, with its hypotheses intact

For the integer-slope ratio U_d , define the periodic step function

$$\Delta(y) = \lfloor dLy \rfloor + \lfloor cy \rfloor - \lfloor dAy \rfloor - \lfloor dBy \rfloor - \lfloor (c + dm)y \rfloor. \quad (64)$$

Balance makes its period equal to 1. We recall the ordinary factorial-ratio criterion, classically due to Landau; see [4, 16].

Lemma 14.1 (Integer-slope criterion). *For U_d in (57), $U_d(n)$ is integral for every $n \geq 0$ if and only if $\Delta(y) \geq 0$ for every $0 \leq y < 1$.*

Proof. If $\Delta \geq 0$, Legendre's formula gives $\nu_p(U_d(n)) = \sum_{k \geq 1} \Delta(n/p^k) \geq 0$ for every prime p . For the converse, suppose Δ is negative somewhere. Right-continuity and its finite set of jumps give a nonempty open interval $I \subset (0, 1)$ on which it is negative. Take an arbitrarily large prime p and an integer n with $n/p \in I$. If p is larger than all integer slopes, every term in the valuation formula with $k \geq 2$ vanishes, since $n < p$. Then $\nu_p(U_d(n)) = \Delta(n/p) < 0$, contrary to integrality. No distribution result for primes is needed; arbitrarily large primes suffice. $\square$

This is Part I's Lemma 2.1 applied to the ordinary ratio U_d ; both Parts state it as the classical criterion and prove it the same way.

Lemma 14.2 (All residue shifts are ordinary slices). *For $j \in \{0, \dots, d-1\}$ and $s \equiv -cj \pmod{d}$, $0 \leq s < d$,*

$$\Delta\left(\frac{j+x}{d}\right) = F_s(x) \quad (0 \leq x < 1). \quad (65)$$

Consequently, $\Delta \geq 0$ on $[0, 1)$ if and only if every $F_s \geq 0$ there.

Proof. Write $cj = da - s$. The three integer-slope floors in the first part of (64) contribute the constant $(L - A - B)j = mj$. The remaining two floors have constants a and $a + mj$, respectively. They cancel the mj , leaving exactly (61). Because $(c, d) = 1$, the map $j \mapsto -cj \pmod{d}$ runs through all residues. $\square$

Combining these two lemmas with Lemma 13.1 proves $\nu_p(V(n)) \geq 0$ for all $p \nmid d$ under the hypothesis of Theorem 12.1.

14.2 The exceptional primes, including the composite case

Let

$$T_n = \frac{(Ln)!}{(An)!(Bn)!} = \binom{Ln}{An, Bn, mn} (mn)! \in \mathbb{Z}.$$

Fix a prime $p \mid d$, write $e = \nu_p(d)$, and for $n > 0$ write $h = \nu_p(n)$. If $h < e$, every $cn + dj$ has valuation exactly h , since $p \nmid c$. Equation (60) therefore gives the exact formula

$$\nu_p(V(n)) = mn(e - h) + \nu_p(T_n) \geq mn(e - h). \quad (66)$$

If $h \geq e$, put $d = p^e d_0$ and $n = p^e N$. Replace the parameter tuple by

$$(L', A', B', m', c', d') = (p^e L, p^e A, p^e B, p^e m, c, d_0). \quad (67)$$

The corresponding V' satisfies $V'(N) = V(n)$ by direct substitution into (56), and its ordinary ratio is exactly the same $U_d(t)$ as before. But $p \nmid d'$, so Lemmas 13.1 and 14.2 apply to the new tuple and give $\nu_p(V(n)) \geq 0$. This includes $d' = 1$, where the only shift is 0.

We have treated every prime. Since $V(n)$ is rational, it is integral. Moreover, (66) proves the divisibility by $\prod_{p \mid d} p^{mn \max(\nu_p(d) - \nu_p(n), 0)}$, which is the factor in (58). The case $n = 0$ is immediate from $V(0) = 1$. This proves the arithmetic part of Theorem 12.1 without restricting d to be prime.

14.3 Small complete certificates for the five cases

For these examples, each F_s takes only the values 0 and 1. The following table gives its complete 1-set; it is zero elsewhere in $[0, 1)$. All intervals are left-closed and right-open.

Sequence	s	$\{x : F_s(x) = 1\}$
A347854	0	$[1/6, 2/3) \cup [5/6, 1)$
	1	$[1/6, 1/3) \cup [5/6, 1)$
A347855	0	$[1/4, 1)$
	1	$[3/4, 1)$
	2	$[1/4, 1/2) \cup [3/4, 1)$
A347856	0	$[1/6, 1/4) \cup [1/3, 3/4) \cup [5/6, 1)$
	1	$[1/6, 1/4) \cup [2/3, 3/4) \cup [5/6, 1)$
A347857	0	$[1/6, 2/5) \cup [2/3, 4/5) \cup [5/6, 1)$
	1	$[1/6, 1/5) \cup [1/3, 3/5) \cup [5/6, 1)$
A347858	0	$[1/9, 1/2) \cup [5/9, 2/3) \cup [7/9, 1)$
	1	$[2/9, 1/3) \cup [4/9, 1/2) \cup [8/9, 1)$

For clarity about its finite nature, a term $\lfloor ax + b \rfloor$ can jump only at $x = (j - b)/a$, $j \in \mathbb{Z}$. Intersect the union of these rational breakpoints with $[0, 1]$ and evaluate at each left endpoint. Between

consecutive breakpoints all five floors are constant. This proves the displayed identities on the entire domain, including the jump points, rather than only on a numerical grid.

The eleven rows require respectively 6, 6, 4, 4, 4, 8, 8, 8, 8, 10, 10 cells. The supplied program regenerates the breakpoints from the parameters and compares the resulting sets with the table. By Lemmas 14.1 and 14.2, the table also proves integrality of all four ordinary bases, independently of their classification. The arithmetic part of Theorem 11.1 now follows.

15 Refinements for the fractional sequences

15.1 Denominator powers in nonzero residue classes

The general theorem immediately gives

$$3^n \mid a_2(n) \quad (3 \nmid n), \quad 2^{m_i n} \mid a_i(n) \quad (n \text{ odd}, i = 1, 3, 4, 5), \quad (68)$$

where $(m_1, m_3, m_4, m_5) = (1, 1, 1, 4)$. In particular, $a_2(3n+1)/3^{3n+1}$ and $a_2(3n+2)/3^{3n+2}$ are integers. These are the integrality conditions in the June 28, 2026 refinement of A347855 [10].

15.2 A linear divisor and an exact binary valuation

Theorem 15.1. *For every $N \geq 1$, $6N - 1 \mid a_1(N)$. For odd N ,*

$$\nu_2(a_1(N)) = 2N + s_2(N), \quad \frac{a_1(N)}{2^{2N+1}(6N-1)} \in \mathbb{Z}, \quad (69)$$

where s_2 is the binary digit sum.

Proof. Let $q = p^k \mid 6N - 1$. Then $p \geq 5$. If $r = N \bmod q$, we have $6r = tq + 1$, where t is either 1 or 5. Indeed, $0 \leq r < q$ and reduction modulo 6 forces t to be a unit modulo 6 in the indicated range. Consequently

$$\frac{r}{q} \in [1/6, 1/3) \text{ or } [5/6, 1).$$

The assertion includes the smallest possibility $q = 5$. Both intervals belong to the 1-set of *both* F_0 and F_1 for a_1 . Thus every level $k \leq \nu_p(6N - 1)$ contributes 1 to (63); all other contributions are nonnegative. This proves the linear divisibility.

When N is odd, all progression factors $N + 2j$ are odd, so

$$\nu_2(a_1(N)) = N + \nu_2((6N)!) - \nu_2((3N)!) - \nu_2((2N)!).$$

Using $\nu_2(M!) = M - s_2(M)$ and $s_2(2M) = s_2(M)$ gives $2N + s_2(N)$. Since $s_2(N) \geq 1$ and $6N - 1$ is odd, the two divisors combine. $\square$

The integer asserted conditionally in the June refinement of A347854 can now be exhibited. Direct cancellation of factorials gives

$$a_1(2n+1) = 2^{4n+3}(12n+5) \frac{12n+1}{(2n+1)(6n+1)} \tilde{B}(n). \quad (70)$$

Taking $N = 2n + 1$ in (69) proves

$$\frac{12n + 1}{(2n + 1)(6n + 1)} \tilde{B}(n) \in \mathbb{Z} \quad (n \geq 0). \quad (71)$$

The identity is recorded in the entry [9]; alternatively it follows from $(j+1/2)! = \sqrt{\pi}(2j+2)!/(4^{j+1}(j+1)!)$ in the Gamma factors. The program checks the identity independently in rational arithmetic.

16 Denominator clearing for the ordinary ratios

This section proves Theorem 11.2. The arguments are useful beyond an isolated divisibility check: they identify the one-sided neighborhoods of rational numbers that account for arbitrarily large prime powers. For A295431 they are a second route to Part I: the step function (73) is Part I's Δ_B on its twelve grid cells, and the covering arguments below are the one-sided grid-interval arguments of Part I's Proposition 5.2 (stated there for A) and Proposition 5.3, here completed at the small primes 2 and 3 by digit sums instead of by Part I's lcm bound or digit-graph certificates.

16.1 Two step functions and two digit identities

For U in (51), its ordinary valuation function is

$$\Delta_U(x) = \lfloor 12x \rfloor + \lfloor x \rfloor - \lfloor 6x \rfloor - \lfloor 4x \rfloor - \lfloor 3x \rfloor. \quad (72)$$

On $[0, 1)$ it is the indicator of

$$[1/12, 1/3) \cup [5/12, 1/2) \cup [7/12, 2/3) \cup [11/12, 1). \quad (73)$$

For $\tilde{B}$, the corresponding function is the indicator of

$$[1/12, 1/6) \cup [5/12, 1/2) \cup [7/12, 5/6) \cup [11/12, 1). \quad (74)$$

Both statements follow by evaluating the twelve cells with endpoints $j/12$. In particular both ordinary ratios are integral. For every prime p ,

$$\nu_p(U(n)) = \sum_{e \geq 1} \Delta_U((n \bmod p^e)/p^e). \quad (75)$$

Legendre's digit formula also gives, for $n \geq 0$,

$$\nu_2(U(n)) = s_2(3n), \quad \nu_3(U(n)) = \frac{1}{2}s_3(2n). \quad (76)$$

For example, the first identity follows by replacing each factorial valuation by its argument minus its binary digit sum; all terms except $s_2(3n)$ cancel. For $\tilde{B}$ the same calculation gives

$$\nu_2(\tilde{B}(n)) = s_2(n), \quad \nu_3(\tilde{B}(n)) = \frac{1}{2}s_3(2n). \quad (77)$$

Here $s_3(2n)$ is even, since ternary digit sum preserves parity.

16.2 The four specified multipliers (second route to Part I)

Second proof of the first four assertions in (52). (Manuscript 01. The first proof, with optimality, is Part I's Section 6.) We first prove $385U(n)/(n+1) \in \mathbb{Z}$. For $p \geq 5$ and $q = p^e \mid n+1$, the residue ratio in (75) is $1 - 1/q$. If $q \geq 13$, it belongs to $[11/12, 1)$, so the corresponding contribution is 1. The only possible uncovered levels are $q = 5, 7, 11$, one for each of those primes. The factor $385 = 5 \cdot 7 \cdot 11$ covers them.

At $p = 2$, put $k = \nu_2(n+1)$. If $k = 0$ there is nothing to prove; if $k = 1$, n is odd and $s_2(3n) \geq 1$. If $k \geq 2$, write $n = 2^k a - 1$ and observe

$$3n = 2^k(3a - 1) + (2^k - 3), \quad s_2(3n) = s_2(3a - 1) + (k - 1) \geq k.$$

At $p = 3$, write $n = 3^k a - 1$, where $k = \nu_3(n+1) \geq 1$. Then

$$2n = 3^k(2a - 1) + (3^k - 2), \quad s_3(2n) \geq 1 + (2k - 1) = 2k.$$

The digit identities prove the required divisibility at 2 and 3.

For $5U(n)/(2n+1)$, the prime 2 is absent. At $p = 3$, if $3^k \mid 2n+1$, the integer $2n$ ends in k ternary digits equal to 2, so (76) supplies at least k factors of 3. For $p \geq 5$, $q = p^e \mid 2n+1$ gives

$$(n \bmod q)/q = \frac{1}{2} - \frac{1}{2q}.$$

This belongs to $[5/12, 1/2)$ for $q \geq 7$. Only $q = 5$ can be uncovered, and the prefactor 5 supplies it.

For $U(n)/(3n+1)$, the prime 3 is absent. If $2^k \mid 3n+1$, then $3n$ ends in k binary ones, so $\nu_2(U(n)) \geq k$. For $p \geq 5$ and $q = p^e \mid 3n+1$, writing $r = n \bmod q$ gives

$$r/q = \frac{1}{3} - \frac{1}{3q} \quad \text{or} \quad r/q = \frac{2}{3} - \frac{1}{3q}.$$

For $q \geq 5$ these belong, respectively, to $[1/12, 1/3)$ and $[7/12, 2/3)$. Every denominator level is therefore supplied by (75).

Finally, $n+1$, $2n+1$, and $3n+1$ are pairwise coprime at every odd prime:

$$\gcd(n+1, 2n+1) = \gcd(2n+1, 3n+1) = 1, \quad \gcd(n+1, 3n+1) \mid 2.$$

The preceding bounds show that the odd-prime deficit for their product is at most one factor each of 5, 7, 11. At 2, only $n+1$ and $3n+1$ can contribute. Their minimum valuation is at most 1, and $U(n)$ covers the larger of the two valuations. One extra factor of 2 suffices. This gives $770 = 2 \cdot 385$. The case $n = 0$ is immediate. $\square$

16.3 Two infinite families with explicit constants (second route to Part I)

Second proof of (53). (Manuscript 01. Part I proves these families with smaller constants in Proposition 5.3 and (26).) Fix $r \geq 1$ and $k \in \{1, 2, 3\}$, and let $E(n) = \prod_{i=1}^r (kn + i)$. First suppose $p \nmid k$ and $q = p^e > 12r$. At most one index $i \in [1, r]$ can satisfy $q \mid kn + i$. If such an index exists, write $n = qa + b$, $0 \leq b < q$. Then

$$kb + i = tq, \quad 1 \leq t \leq k, \quad \frac{b}{q} = \frac{t}{k} - \frac{i}{kq}.$$

For $k = 1, 2, 3$, every t/k has a left neighborhood of width at least $1/12$ in the 1-set (73). Since $i/(kq) < 1/12$, we have $\Delta_U(b/q) = 1$. Therefore every denominator contribution at a level $q > 12r$ is covered by the corresponding term of (75). At each of the remaining levels, the denominator count is at most r . Consequently

$$\nu_p(E(n)) - \nu_p(U(n)) \leq r \left\lfloor \log_p(12r) \right\rfloor \quad (p \nmid k). \quad (78)$$

The right side is precisely the valuation of L_{12r}^r .

When $p \mid k$, either $k = 2 = p$ or $k = 3 = p$. Only indices $i = pj$ contribute at this prime. With $s = \lfloor r/p \rfloor$,

$$\nu_p(E(n)) = s + \nu_p \left(\prod_{j=1}^s (n+j) \right).$$

The already proved $k = 1$ case bounds the deficit of the product by $s \lfloor \log_p(12s) \rfloor$ if $s > 0$; for $s = 0$ it is zero. This is bounded by the valuation of $k^r L_{12r}^r$. Thus $C(k, r)$ clears every prime, proving the first family.

For the second family put

$$E_-(n) = \prod_{\substack{1 \leq i \leq r \\ (i, 12)=1}} (12n - i).$$

No factor is zero, and none is divisible by 2 or 3. Let $p \geq 5$ and $q = p^e > r$. Again, at most one index is counted at level q . If $q \mid 12n - i$ and $b = n \bmod q$, then

$$12b - i = tq, \quad t \in \{1, 5, 7, 11\}, \quad \frac{b}{q} = \frac{t}{12} + \frac{i}{12q}.$$

To see the range of t , use $q > i$, $0 \leq b < q$, and reduction modulo 12; $t = 0$ is excluded by $(i, 12) = 1$. Each of the four values $t/12$ has a right neighborhood of width at least $1/12$ contained in (73). Thus the level is covered by Δ_U . The remaining levels $q \leq r$ contribute a deficit of at most r each. Their total is at most $r \lfloor \log_p r \rfloor = \nu_p(L_r^r)$. This proves the second family, including negative factors when n is small, since valuations concern their absolute values. $\square$

For $r = 1$, $D(1) = 1$, so the second family gives $U(n)/(12n - 1) \in \mathbb{Z}$. This also follows by setting $N = 2n$ in Theorem 15.1. Thus the fifth assertion in (52) is proved in two compatible ways.

16.4 The product divisor in A295432

Proof of (54). Equation (77) shows $6 \mid \tilde{B}(n)$ for $n \geq 1$. Both $6n + 1$ and $12n - 1$ are coprime to 6, and their greatest common divisor divides 3 and is therefore 1.

If $q = p^e \mid 6n + 1$, then $p \geq 5$ and the residue ratio is

$$(n \bmod q)/q = \frac{1}{6} - \frac{1}{6q} \quad \text{or} \quad \frac{5}{6} - \frac{1}{6q}.$$

For $q \geq 5$, both lie in the 1-set (74). If instead $q = p^e \mid 12n - 1$, the residue ratio is

$$\frac{t}{12} + \frac{1}{12q}, \quad t \in \{1, 5, 7, 11\}.$$

These four points also lie in (74), including at $q = 5$. The ordinary valuation formula for $\tilde{B}$ therefore covers every level of both linear factors. The coprimality noted above let us multiply all three divisors, proving the assertion. $\square$

17 Algebraicity by fractional Frobenius solutions

17.1 The hypergeometric equations

For a balanced ordinary factorial ratio

$$W(t) = \frac{\prod_{a \in \mathcal{A}} \Gamma(1 + at)}{\prod_{b \in \mathcal{B}} \Gamma(1 + bt)},$$

expand the fractions j/a , $1 \leq j \leq a$, for all numerator slopes and the analogous denominator fractions, then cancel common fractions with multiplicity. In the height-one cases here, the remaining lists $\alpha_1, \dots, \alpha_h$ and $\beta_1, \dots, \beta_h$ have equal length, with 1 included among the β 's. Direct use of the Gamma functional equation gives

$$\frac{W(t+1)}{W(t)} = \lambda \frac{\prod_{j=1}^h (t + \alpha_j)}{\prod_{j=1}^h (t + \beta_j)}, \quad \lambda = \frac{\prod_{a \in \mathcal{A}} a^a}{\prod_{b \in \mathcal{B}} b^b}. \quad (79)$$

For the four bases, the complete lists are as follows. They agree with the ordinary-ratio parameters recorded in the OEIS cross-reference table [14], and can also be obtained by the cancellation just described.

Base	α	β (including 1)	λ
I	$\frac{1}{12}, \frac{5}{12}, \frac{7}{12}, \frac{11}{12}$	$\frac{1}{3}, \frac{1}{2}, \frac{2}{3}, 1$	27648
II	$\frac{1}{12}, \frac{1}{6}, \frac{5}{12}, \frac{7}{12}, \frac{5}{6}, \frac{11}{12}$	$\frac{1}{8}, \frac{3}{8}, \frac{1}{2}, \frac{5}{8}, \frac{7}{8}, 1$	19683/4
III	$\frac{1}{12}, \frac{1}{3}, \frac{5}{12}, \frac{7}{12}, \frac{2}{3}, \frac{11}{12}$	$\frac{1}{5}, \frac{2}{5}, \frac{1}{2}, \frac{3}{5}, \frac{4}{5}, 1$	$2^{10}3^9/5^5$
IV	$\frac{1}{18}, \frac{5}{18}, \frac{7}{18}, \frac{11}{18}, \frac{13}{18}, \frac{17}{18}$	$\frac{1}{4}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, 1$	2^43^{12}

Let $\theta = z d/dz$ and define

$$Q(T) = \prod_j (T + \beta_j - 1), \quad P(T) = \prod_j (T + \alpha_j), \quad \mathcal{L} = Q(\theta) - \lambda z P(\theta). \quad (80)$$

These are generalized hypergeometric equations with regular singularities only at 0, $1/\lambda$, and infinity. (Part I carries out the same cancellation for A211417 and its normalized companion, in (34) and (40); it uses no monodromy.)

17.2 The classical input and what the certificates replace

We use the following classical monodromy facts, not new results of this paper. For disjoint exponent sets $e^{2\pi i \alpha_j}$ and $e^{2\pi i \beta_j}$, the monodromy group of (80) is conjugate to the group generated by the companion matrices of

$$f(T) = \prod_j (T - e^{2\pi i \alpha_j}), \quad g(T) = \prod_j (T - e^{2\pi i \beta_j}). \quad (81)$$

This is Levelt's description, as presented in Beukers–Heckman [17, Section 3]. A directly accessible presentation is [18, Theorem 3.2.3 and Section 3.3]. Strict interlacing of the roots on the unit circle gives a positive-definite invariant Hermitian form; see [17, Section 4].

For our four cases, Section 18 supplies that form as an explicit integer matrix. Thus their proof needs the companion description but does not require recognizing a group in the finite-group classification or trusting a table of group orders. The step from positive definiteness and integrality to finiteness is proved below.

Lemma 17.1 (Integral isometries of a definite form). *If a positive-definite Hermitian matrix H is preserved by a subgroup $G \subset \mathrm{GL}_h(\mathbb{Z})$, then G is finite.*

Proof. Let $\eta > 0$ be the least eigenvalue of H . For each matrix $M \in G$, its j th column v_j satisfies

$$\eta \|v_j\|^2 \leq v_j^* H v_j = H_{jj}.$$

There are only finitely many integer vectors satisfying this bound. There are therefore only finitely many possible ordered lists of columns. $\square$

Lemma 17.2 (Finite monodromy and regular singularities). *Every local solution of a regular-singular linear differential equation on the Riemann sphere with finite monodromy is algebraic over $\mathbb{C}(z)$.*

Proof. A local solution has only finitely many continuations because its monodromy orbit is finite. Pass to the finite covering that trivializes the monodromy. At the finitely many points over singularities, regular singularity gives at most power-logarithmic growth. Finite local monodromy removes logarithmic terms and, after a finite ramification, makes all local exponents integral. Thus the continued solution is meromorphic on the compactified finite cover. Equivalently, the elementary symmetric functions of its finitely many branches are single-valued meromorphic functions on the sphere, hence rational. The monic polynomial with those branches as roots proves algebraicity. $\square$

Regular singularity is indispensable in this statement: finite monodromy alone would not rule out an essential singularity.

17.3 Why the nonintegral subsequences solve the same equation

Lemma 17.3 (Fractional Frobenius sampling). *Suppose W satisfies (79). If $s \geq 0$ and $Q(s) = 0$, then*

$$\Phi_s(z) = \sum_{k \geq 0} W(k+s) z^{k+s} \quad (82)$$

is a local Puiseux solution of $\mathcal{L}\Phi_s = 0$, on a chosen branch of z^s .

Proof. The coefficient of z^s in $\mathcal{L}\Phi_s$ is $Q(s)W(s) = 0$. For $k \geq 1$, the coefficient of z^{k+s} is

$$Q(k+s)W(k+s) - \lambda P(k-1+s)W(k-1+s),$$

which vanishes by (79). The series converges near zero by the ratio test, since its consecutive coefficient ratio tends to λ . $\square$

This boundary term is important. Merely shifting the parameters in a hypergeometric series does not, by itself, prove that the shifted series is another solution of the original homogeneous equation.

For each row of the five-sequence parameter table, the list of β 's contains

$$1, 1 - 1/d, \dots, 1 - (d-1)/d.$$

Hence $Q(r/d) = 0$ for $0 \leq r < d$. If the base equation has finite monodromy, all the corresponding $\Phi_{r/d}$ are algebraic, and

$$\sum_{n \geq 0} V(n)x^n = \sum_{r=0}^{d-1} \Phi_{r/d}(x^d). \quad (83)$$

The identity holds as a power series after choosing the Puiseux branches with leading term $W(r/d)z^{r/d}$. Each summand contributes exactly the terms with $n \equiv r \pmod{d}$.

Lemma 17.4 (Descent of algebraicity). *A series in $\mathbb{Q}[[x]]$ that is algebraic over $\mathbb{C}(x)$ is algebraic over $\mathbb{Q}(x)$.*

Proof. A complex polynomial relation of bounded degree in x and the series gives a nonzero solution of a homogeneous linear system whose entries are rational: set each coefficient of the substituted series equal to zero. There are finitely many unknown polynomial coefficients, so the rational row space has a finite basis even though the system initially has infinitely many rows. A nontrivial complex kernel implies a nontrivial rational kernel for that finite rational matrix. The resulting rational coefficients give a nonzero relation over $\mathbb{Q}(x)$. $\square$

Equation (60) supplies rational coefficients, so the descent lemma applies to (83).

17.4 Completion of the general algebraicity theorem

For the general U_d of (57), integrality and Lemma 14.1 imply $\Delta \geq 0$. Away from the finitely many jump points, height one and balance give

$$\Delta(y) + \Delta(1 - y) = 1.$$

Consequently Δ takes only the values 0 and 1, first away from jumps and then everywhere by right-continuity. After common numerator and denominator fractions are canceled, the jumps are the α 's and β 's, respectively. A jump of size greater than one is impossible. Thus all remaining roots are simple and strictly alternate, beginning with an α and ending with the denominator root at 1.

The polynomials f, g are obtained by canceling common factors in

$$\frac{(T^{dL} - 1)(T^c - 1)}{(T^{dA} - 1)(T^{dB} - 1)(T^{c+dm} - 1)}.$$

They are monic integral products of cyclotomic polynomials with constant terms ± 1 . Interlacing, the classical invariant-form result, and Lemma 17.1 imply finite monodromy.

It remains to justify all required exponents, rather than only $s = 0$. For $1 \leq j < d$, the fraction j/d occurs once in the fraction list for dL , once for dA , and once for dB . It occurs in neither the c list nor the $c + dm$ list, since both are coprime to d . Cancellation therefore leaves one denominator occurrence. Height one leaves one denominator occurrence of 1 as well. Hence every r/d , $0 \leq r < d$, is a zero of Q . Now Lemmas 17.2 to 17.4 and (83) prove the algebraicity part of Theorem 12.1.

18 Four exact Gram-matrix certificates

For a monic polynomial $f(T) = T^h + f_{h-1}T^{h-1} + \cdots + f_0$, our companion matrix has ones on the subdiagonal and last column $(-f_0, -f_1, \dots, -f_{h-1})^\top$. Denote it by C_f . For the four base equations, the polynomial pairs are

$$\begin{aligned} f_{\text{I}} &= T^4 - T^2 + 1, & g_{\text{I}} &= T^4 + T^3 - T - 1, \\ f_{\text{II}} &= T^6 - T^5 + T^3 - T + 1, & g_{\text{II}} &= T^6 - T^4 + T^2 - 1, \\ f_{\text{III}} &= T^6 + T^5 - T^3 + T + 1, & g_{\text{III}} &= T^6 + T^5 - T - 1, \\ f_{\text{IV}} &= T^6 - T^3 + 1, & g_{\text{IV}} &= T^6 + T^5 + T^4 - T^2 - T - 1. \end{aligned} \tag{84}$$

Equivalently, in terms of cyclotomic polynomials,

Base	f	g
I	Φ_{12}	$(T^2 - 1)\Phi_3$
II	$\Phi_{12}\Phi_6$	$(T^2 - 1)\Phi_8$
III	$\Phi_{12}\Phi_3$	$(T^2 - 1)\Phi_5$
IV	Φ_{18}	$(T^2 - 1)\Phi_3\Phi_4$

The pairs have no common root and their companion matrices are unimodular. Here are invariant forms:

$$H_{\text{I}} = \begin{pmatrix} 2 & -1 & 1 & 0 \\ -1 & 2 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 0 & 1 & -1 & 2 \end{pmatrix}, \tag{85}$$

$$H_{\text{II}} = \begin{pmatrix} 4 & 2 & 1 & -1 & -2 & -1 \\ 2 & 4 & 2 & 1 & -1 & -2 \\ 1 & 2 & 4 & 2 & 1 & -1 \\ -1 & 1 & 2 & 4 & 2 & 1 \\ -2 & -1 & 1 & 2 & 4 & 2 \\ -1 & -2 & -1 & 1 & 2 & 4 \end{pmatrix}, \tag{86}$$

$$H_{\text{III}} = \begin{pmatrix} 10 & -8 & 4 & 1 & -5 & 7 \\ -8 & 10 & -8 & 4 & 1 & -5 \\ 4 & -8 & 10 & -8 & 4 & 1 \\ 1 & 4 & -8 & 10 & -8 & 4 \\ -5 & 1 & 4 & -8 & 10 & -8 \\ 7 & -5 & 1 & 4 & -8 & 10 \end{pmatrix}, \tag{87}$$

$$H_{\text{IV}} = \begin{pmatrix} 4 & -1 & -2 & 2 & 1 & -1 \\ -1 & 4 & -1 & -2 & 2 & 1 \\ -2 & -1 & 4 & -1 & -2 & 2 \\ 2 & -2 & -1 & 4 & -1 & -2 \\ 1 & 2 & -2 & -1 & 4 & -1 \\ -1 & 1 & 2 & -2 & -1 & 4 \end{pmatrix}. \tag{88}$$

Their leading principal minors are, respectively,

$$\begin{array}{l|l} \text{I} & 2, 3, 4, 4 \\ \text{II} & 4, 12, 36, 81, 162, 243 \\ \text{III} & 10, 36, 72, 108, 162, 243 \\ \text{IV} & 4, 15, 36, 81, 162, 243. \end{array} \quad (89)$$

All are positive, so Sylvester's criterion proves positive definiteness. Exact multiplication gives, for each base J ,

$$C_{f_J}^\top H_J C_{f_J} = H_J = C_{g_J}^\top H_J C_{g_J}. \quad (90)$$

These are finite integer identities, not approximate numerical eigentests. Their validity can be checked directly from the printed data. The program also derives the reduced polynomial pair by exact polynomial greatest-common-divisor cancellation from the original factorial slopes, which prevents an unnoticed mismatch between the differential equation and the matrix certificate.

By Lemma 17.1, the four companion groups are finite. By Levelt's identification and Lemma 17.2, all solutions of the four equations are algebraic. The required roots $0, 1/2$ for the half-integer cases, and $0, 1/3, 2/3$ for the third-integer case, are present in Q . Applying (83) proves the algebraicity assertion of Theorem 11.1. This completes the proof of that theorem.

Remark 18.1 (Independent certificate routes). The program also checks Galois-stable interlacing of the root sets for moduli 12, 24, 60, 36, respectively. This is an additional check, not a necessary step in the Gram-matrix proof. The matrices establish finiteness without computing the order of any generated group, and no such order is asserted.

19 Positive moments, Hankel determinants, and shape

The following consequences explain the regular behavior of these sequences. Moment representations for some ordinary and fractional members are already recorded in the OEIS material, notably by Penson [14, 9]. The beta-product construction gives a uniform proof for all five cases.

Let α_j, β_j be listed in increasing order in the parameter table. They satisfy $0 < \alpha_j < \beta_j$. Let Z_j be independent beta random variables with density

$$\frac{z^{\alpha_j-1}(1-z)^{\beta_j-\alpha_j-1}}{B(\alpha_j, \beta_j - \alpha_j)} dz, \quad 0 < z < 1, \quad (91)$$

and set

$$Y = \lambda \prod_{j=1}^h Z_j, \quad X = Y^{1/d}, \quad R = \lambda^{1/d}. \quad (92)$$

Theorem 19.1 (A compactly supported positive moment law). *For the relevant base and denominator d , the corresponding V satisfies*

$$V(t) = \mathbb{E}[X^t] \quad (t \geq 0). \quad (93)$$

The law μ of X has support $[0, R]$ and assigns positive measure to every nonempty open subinterval of $(0, R)$. In particular, for every $\ell \geq 0$ and $q \geq 1$,

$$\det[V(\ell + i + j)]_{0 \leq i, j < q} > 0. \quad (94)$$

Also, the normalized sequence $b_n = V(n)/R^n$ has strictly positive alternating finite differences:

$$(-1)^k \Delta^k b_n > 0 \quad (n, k \geq 0). \quad (95)$$

Proof. The beta integral gives

$$\mathbb{E}[Z_j^t] = \frac{\Gamma(\alpha_j + t)\Gamma(\beta_j)}{\Gamma(\alpha_j)\Gamma(\beta_j + t)}.$$

Gauss's multiplication formula, with the common fractions canceled and the normalization $U_J(0) = 1$, yields

$$U_J(t) = \lambda^t \prod_j \frac{\Gamma(\alpha_j + t)\Gamma(\beta_j)}{\Gamma(\alpha_j)\Gamma(\beta_j + t)}.$$

Independence proves $\mathbb{E}[Y^t] = U_J(t)$ and hence $\mathbb{E}[X^t] = U_J(t/d) = V(t)$. The beta densities are positive throughout $(0, 1)$, so the product and its positive root have the stated support property.

For a nonzero real polynomial P of degree less than q ,

$$\int_0^R x^\ell P(x)^2 d\mu(x) > 0.$$

Indeed, P has only finitely many zeros, and μ is not supported on them. Thus the Hankel matrix is a positive-definite Gram matrix, proving (94). Finally, with $T = X/R \in (0, 1)$ almost surely,

$$(-1)^k \Delta^k b_n = \mathbb{E}[T^n(1 - T)^k] > 0.$$

□

In particular, strict Cauchy–Schwarz gives

$$V(n+1)^2 < V(n)V(n+2) \quad (n \geq 0), \quad (96)$$

so the consecutive ratios are strictly increasing. All five first ratios are greater than 1; therefore each sequence is strictly increasing. The ratio limit is R , either by the asymptotics below or directly from the moment law and its upper endpoint. The moment problem on $[0, R]$ is determinate: equality of all moments gives equality of integrals of polynomials, then of continuous functions by polynomial approximation, and therefore equality of the measures. No uniqueness claim on a larger class of signed distributions is intended.

20 All-orders asymptotics and effective errors

The leading growth scale follows from Stirling's formula, which already underlies the asymptotic information in the entries. We give the complete coefficient rule and explicit remainder control. Let $t = c/d$ in this section (the letter t is a fixed slope, not the running argument), and define

$$R = \frac{L^L t^t}{A^A B^B (t+m)^{t+m}}, \quad K = \frac{1}{\sqrt{2\pi}} \sqrt{\frac{Lt}{AB(t+m)}}. \quad (97)$$

For $j \geq 1$, put

$$\gamma_{2j-1} = \frac{B_{2j}}{2j(2j-1)} \left(L^{1-2j} + t^{1-2j} - A^{1-2j} - B^{1-2j} - (t+m)^{1-2j} \right), \quad (98)$$

where B_{2j} is a Bernoulli number, and set $\gamma_{2j} = 0$.

Theorem 20.1 (An effective all-orders expansion). *For any fixed integer $J \geq 1$, as $n \rightarrow +\infty$ through real values,*

$$\log V(n) = n \log R - \frac{1}{2} \log n + \log K + \sum_{j=1}^J \frac{\gamma_{2j-1}}{n^{2j-1}} + \mathcal{R}_J(n), \quad (99)$$

where for every $n > 0$,

$$|\mathcal{R}_J(n)| \leq E_J n^{-(2J+1)}, \quad (100)$$

$$E_J = \frac{|B_{2J+2}|}{(2J+2)(2J+1)} \left(L^{-2J-1} + t^{-2J-1} + A^{-2J-1} + B^{-2J-1} + (t+m)^{-2J-1} \right).$$

In particular,

$$V(n) \sim K R^n n^{-1/2} \exp \left(\sum_{j \geq 1} \gamma_{2j-1} n^{-(2j-1)} \right). \quad (101)$$

Proof. For $a > 0$, substitute $z = an$ into the logarithmic Stirling expansion for $\Gamma(z)$, and add $\log(an)$ to obtain the expansion for $\Gamma(1+an)$. On the positive axis the remainder after a fixed number of terms is bounded in magnitude by the first omitted term; see [8, Section 5.11]. Combining the two numerator and three denominator expansions, the $n \log n$ and linear-in- n terms cancel by balance. The surviving exponential constant is R , the power is $n^{-1/2}$, and the constant is K . Each odd inverse-power coefficient is (98). Adding the absolute bounds on the five remainders gives (100). $\square$

[write] **Added 1 October 2026, batch 74: an instance of repository results.** Apart from the explicit constant E_J , Theorem 20.1 is an instance of expansions already in the repository. Part I's (42) with numerator slopes (L, t) , denominator slopes $(A, B, t+m)$ and height $h = 1$ gives (99) with $\Lambda = R$, $K_R = K$ and $\sigma_{2j-1} = \gamma_{2j-1}$; its derivation from DLMF 5.11 does not use integrality of the slopes. It is also the balanced case $\kappa = 0$, $\lambda = \log R$, $\beta = -\frac{1}{2}$ of the gamma-quotient coefficient engine `t2:thm:gamma` of the combinatorial-inverses volume [21], with $q_j = \gamma_j$. New here is the fixed-order bound (100) with its constant E_J and its evaluation at the rational slope $t = c/d$.

The first constants are displayed below. The column headed $K\sqrt{\pi}$ keeps the table compact.

i	R_i	$K_i\sqrt{\pi}$	γ_1	γ_3
1	$96\sqrt{3}$	$1/\sqrt{6}$	$1/18$	$-163/7776$
2	$24\sqrt[3]{2}$	$1/2$	$1/12$	$-163/2304$
3	$81\sqrt{3}/2$	$1/2$	$1/48$	$-257/13824$
4	$2592\sqrt{15}/125$	$\sqrt{30}/10$	$-1/30$	$-337/1620000$
5	2916	$1/\sqrt{12}$	$19/216$	$-9131/419904$

For an ordinary inverse-power expansion

$$V(n) \sim K R^n n^{-1/2} \sum_{k \geq 0} c_k n^{-k}, \quad c_0 = 1,$$

all coefficients follow recursively from

$$k c_k = \sum_{j=1}^k j \gamma_j c_{k-j}. \quad (102)$$

For example,

$$c_1 = \gamma_1, \quad c_2 = \frac{\gamma_1^2}{2}, \quad c_3 = \gamma_3 + \frac{\gamma_1^3}{6}, \quad c_4 = \gamma_1\gamma_3 + \frac{\gamma_1^4}{24}.$$

This is a full Poincaré expansion of the dominant exponential sector. We do not claim that it specifies exponentially small sectors of a resurgent transseries, nor that the formal Stirling series converges. The effective fixed-order bound (100) is the assertion needed for numerical use.

Corollary 20.2 (The generating functions are genuinely nonrational). *Each A_i has radius of convergence $1/R_i$ and, along the real axis,*

$$A_i(x) \sim K_i \sqrt{\pi} (1 - R_i x)^{-1/2} \quad (x \uparrow 1/R_i). \quad (103)$$

Consequently $A_i(x)$ is not rational over $\mathbb{C}(x)$.

Proof. The radius follows from the coefficient growth. Put $R_i x = e^{-u}$ with $u \downarrow 0$. A Riemann-sum comparison gives $\sum_{n \geq 1} n^{-1/2} e^{-un} \sim \sqrt{\pi} u^{-1/2}$. The coefficient asymptotic and positivity allow comparison above and below outside a fixed finite initial segment, proving (103). Since $u \sim 1 - R_i x$, the stated form follows. A rational function at a finite point can have only an integral-order pole or a removable singularity, not the half-order divergence in (103). $\square$

21 Inverse expansions and index recovery

Fix one of the five continuous Gamma interpolations $V(x)$ and put $\tau = \log R > 0$. The moment representation implies strict convexity of $\log V(x)$: its second derivative is the variance of $\log X$ under the probability law tilted by X^x . In particular, for $x \geq 1$ its derivative is at least the positive secant slope $\log V(1) - \log V(0)$. Thus, for $y \geq V(1)$, there is a unique $x \geq 1$ satisfying $V(x) = y$. Denote this inverse by $\mathcal{I}(y)$.

The leading model $y = KR^x/\sqrt{x}$ has a useful exact inverse on its large branch. Set

$$N(y) = -\frac{1}{2\tau} W_{-1} \left(-\frac{2\tau K^2}{y^2} \right), \quad (104)$$

for sufficiently large y . Here W_{-1} is the real branch with values at most -1 . The branch choice is fixed by the requirement $N(y) \rightarrow \infty$. It satisfies

$$\tau N - \frac{1}{2} \log N = \log(y/K).$$

Theorem 21.1 (Inverse expansion to every order). *As $y \rightarrow \infty$,*

$$\begin{aligned} \mathcal{I}(y) = N - \frac{\gamma_1}{\tau N} - \frac{\gamma_1}{2\tau^2 N^2} \\ - \left(\frac{\gamma_3}{\tau} + \frac{\gamma_1^2}{\tau^2} + \frac{\gamma_1}{4\tau^3} \right) \frac{1}{N^3} + O(N^{-4}), \end{aligned} \quad (105)$$

where $N = N(y)$. There is a recursively computable expansion $\mathcal{I}(y) \sim N + \sum_{j \geq 1} b_j N^{-j}$ to arbitrary fixed order.

Proof. Let $\varepsilon = 1/N$ and write $x = N + s$, where $s = \sum_{j \geq 1} b_j \varepsilon^j$. Subtract the equation defining N from (99). The formal equation for s is

$$\tau s - \frac{1}{2} \log(1 + \varepsilon s) + \sum_{j \geq 1} \gamma_j \varepsilon^j (1 + \varepsilon s)^{-j} = 0. \quad (106)$$

At order ε^j , the only occurrence of the new coefficient b_j is τb_j . Hence b_j is uniquely determined by earlier coefficients. The first equations give

$$b_1 = -\gamma_1/\tau, \quad b_2 = -\gamma_1/(2\tau^2), \quad b_3 = -\gamma_3/\tau - \gamma_1^2/\tau^2 - \gamma_1/(4\tau^3).$$

For an analytic justification at each fixed order, substitute the truncated series into the effective expansion of $\log V$. Its residual has the order of the first omitted power. Since $(\log V)'$ stays bounded away from zero for sufficiently large arguments, the mean value theorem converts that residual to an error of the same order in x . This proves the asymptotic inverse expansion, not merely formal reversion. $\square$

[write] **Added 1 October 2026, batch 74: an instance of repository results.** Theorem 21.1 is the balanced inverse theorem `t2:thm:balanced-inverse` of the combinatorial-inverses volume [21] with $a = \tau = \log R$, $\beta = -\frac{1}{2}$, $C = K$, $q_j = \gamma_j$ and $X = N$. That theorem's coefficients $d_1 = -q_1/a$, $d_2 = \beta q_1/a^2 - q_2/a$ and $d_3 = -\beta^2 q_1/a^3 + (\beta q_2 - q_1^2)/a^2 - q_3/a$ reduce, with $q_2 = \gamma_2 = 0$, to the b_1, b_2, b_3 above, and it supplies a nonrecursive formula for every b_j . The choice of W_{-1} is the branch rule stated there for $\beta < 0$. The index-recovery identity below is part (1), and the advice on certified rounding is the separation condition of part (2), of the staircase theorem `p0:thm:staircase` of the transseries-and-inversion volume [22]. No novelty is claimed for this section.

The Lambert expression can itself be expanded in ordinary logarithms. With

$$Y = 2 \log(y/K) - \log(2\tau), \quad H = \log Y,$$

substitution into $2\tau N - \log(2\tau N) = Y$ gives

$$2\tau N = Y + H + \frac{H}{Y} + \frac{H - H^2/2}{Y^2} + O\left(\frac{H^3}{Y^3}\right). \quad (107)$$

Together with (106), this produces the inverse in a logarithmic asymptotic scale without leaving an unevaluated special function.

For integer index recovery, strict increase of $a_i(n)$ implies the exact identity

$$\min\{n \geq 1 : a_i(n) \geq y\} = \lceil \mathcal{I}_i(y) \rceil \quad (y \geq a_i(1)).$$

An approximate inverse should not be rounded blindly: near an integer, a certified error interval must be used, or the adjacent exact terms must be compared. The logarithmic remainder bound and monotonicity give a direct route to such certification.

22 Reproducibility and the trust boundary

The accompanying file `verify.py` (shipped as `code/02-fractional-dilation-verify.py`; see Appendix D for how to rerun it under its delivery name) uses only the Python standard library, integer arithmetic, and exact rational arithmetic. It needs no network access and no computer-algebra system. Run

```
python verify.py --max-n 200 --output results.txt
```

from a normal Python environment. The supplied execution completed successfully. It checks the eleven complete floor partitions, the four reduced cyclotomic polynomial pairs, eight Gram invariance identities, and all leading principal minors. It also checks the hypergeometric parameter cancellation, the required Frobenius exponents, and Galois-stable interlacing.

Additional exact cross-checks include all five sequences through $n = 200$, the displayed OEIS prefixes, the June refinements, the specified clearing multipliers, and the product divisor for $\tilde{B}$. There are 6120 ordinary prime-valuation comparisons for the target sequences and 9200 comparisons of the general valuation identities, including composite denominators up to 12. The two infinite clearing families are tested for $1 \leq r \leq 8$ and $0 \leq n \leq 30$.

These assertions have different logical roles. The complete floor partition is a finite proof certificate because Lemmas 13.1 and 14.2 reduce the universal arithmetic question to those exact intervals. The Gram identities are finite proof certificates because Lemma 17.1 and the cited companion theorem supply the universal implication. By contrast, checking 201 terms or 9200 valuation identities is a regression test; it cannot prove the assertions for unbounded n, d , or r .

The executable is not a proof-assistant kernel, and its agreement with the manuscript is not independent peer review. The package contains neither a Lean file with placeholders nor an unexecuted claim of formal verification. The arithmetic proofs are designed to admit later formalization. The analytic monodromy bridge is a separate, substantially larger dependency.

For asymptotics, the formulas are symbolic consequences of Stirling's formula and are proved in the text. The coefficient recurrence and inverse construction do not depend on fitting numerical data. The supplementary `asymptotic_demo.py` (shipped as `code/02-fractional-dilation-asymptotic_demo.py`), when the optional `mpmath` dependency is available, evaluates forward and inverse approximations as an additional numerical check; it is not part of the exact proof certificate.

23 Further research questions and topics

23.1 Minimal denominator-clearing constants

The explicit constants $(kL_{12r})^r$ and L_r^r solve the existence questions but are very large. Determine the least positive $C(k, r)$ and $D(r)$, or prove upper bounds with substantially smaller prime support and exponents. The proofs isolate the only potentially deficient prime-power levels, making this a finite local optimization problem for each fixed r . A complete algorithm must account for simultaneous residue constraints across levels, rather than optimize each level independently.

[write] **Added 1 October 2026, batch 74: re-scoped.** Part I already has such an algorithm. Its finite weighted digit graphs (Section 6) couple the residues at $p, p^2, \dots$ exactly and determine the least multipliers for the four specified products, so those cases are settled. Part I also proves smaller family constants than $C(k, r)$ and $D(r)$ (Section 10.2), and it lists the behaviour of the optimal constants as r grows, together with uniform control of the state space in r , as open (Section 9.3). The open question is that growing- r problem.

23.2 Beyond two numerator and three denominator Gamma factors

The rational-dilation theorem covers composite denominators already, under its stated structural hypotheses. The next question is not simply to allow a composite d , but to allow several fractional

Gamma pairs, shared denominator primes, or a more general integer-factorial prefactor. Which structural conditions ensure that all exceptional-prime contributions remain nonnegative? The cancellation step must precede any valuation criterion.

23.3 All rational dilations of the sporadic family

For every sporadic ordinary height-one ratio, classify rational $q > 0$ for which its Gamma interpolation $U(qn)$ is integral for every integer n . Some instances fit Theorem 12.1 after permuting slopes or canceling common factors; others may not. A useful deliverable would be a finite classification of denominator patterns with exact certificates, distinguishing integrality from mere boundedness of denominators.

23.4 Explicit algebraic equations and branch degrees

The proof establishes algebraicity but does not produce minimal polynomials for the five A_i . Compute them, determine their degrees, and relate the degree of the branch sum (83) to its stabilizer in the finite monodromy group. A polynomial guessed from initial coefficients requires an independent identity proof, for example by the differential equation and a sufficient initial-value argument on a specified branch.

23.5 Combinatorial interpretations

Find natural counting models for the fractional sequences and for the normalized integers in (52), (71), and (54). Integrality by valuations and positivity by moments are not, by themselves, combinatorial interpretations. A satisfactory model would explain the denominator-clearing multipliers and the algebraic structure rather than merely encode the resulting integers artificially.

23.6 The remaining supercongruences

The OEIS entries include stronger prime-power congruences that are untouched by the present proof. The fact that each local floor contribution is 0 or 1 controls divisibility, not the unit part of a factorial ratio. A separate analysis of the unit factors, or a suitable p -adic Gamma identity, is needed. This is a natural next arithmetic target, but no supercongruence resolution is claimed here.

23.7 Hankel asymptotics and orthogonal polynomials

The beta-product laws imply positive Hankel determinants but do not give closed forms for them. Determine the large-size Hankel determinant asymptotics, the recurrence coefficients of the corresponding orthogonal polynomials, and the endpoint behavior of the moment densities. The algebraic Stieltjes transforms may provide an additional tool, but the density need not be a simple classical weight.

23.8 Effective algebraic and asymptotic computation

Develop a certified pipeline from a factorial ratio to its floor partition, finite-monodromy certificate, all-orders asymptotics, and inverse index bounds. The logarithmic remainder bound already gives

guaranteed forward bounds. The remaining engineering question is how to select truncation orders and neighboring integer tests efficiently over a wide range of input sizes without confusing fixed-order asymptotics with optimal truncation.

23.9 A formalization plan for ProveIt

The most accessible first module would prove (60), the progression-counting lemma, and the ordinary slice identity over exact integers and rationals. A second module could verify floor partitions and formalize the denominator-prime reduction (67), yielding a reusable arithmetic theorem rather than five isolated proofs. The specified clearing factors and the two infinite families then use only integer valuations, digit sums, and interval arithmetic.

For the analytic branch, the small matrix identities and positive-definiteness certificates are accessible independently of the differential equations. Connecting them to algebraicity requires a formal treatment of monodromy, regular singularities, and Levelt's theorem, or a different route through explicit algebraic equations. Until that connection is formalized, a checked matrix certificate should not be described as a kernel-checked proof of the full generating-function theorem.

A Admissible denominators for all 52 sporadic ratios

In the following table, a and b are the numerator and denominator parameter lists of $R(n)$. The column K_- contains the k with $J(k) = -1$ and therefore describes primitive factors $kn + i$, $i > 0$, $(i, k) = 1$. The column K_+ contains $J(k) = +1$ and describes $kn - i$. The signs refer to jumps, not to the sign of the offset. Every allowed root must still be simple. The parameter data are transcribed from Coserea's table [7]; the two K columns are calculations from Theorem 4.1. Row 1 is B and row 31 is A .

Index	a	b	K_-	K_+
1	12, 1	6, 4, 3	1, 2, 3	12
2	12, 3, 2	6, 6, 4, 1	1, 2, 6	12
3	12, 1	8, 3, 2	1, 2, 8	6, 12
4	12, 3	8, 6, 1	1, 2, 8	3, 12
5	12, 3	6, 5, 4	1, 2, 5	3, 12
6	12, 5	10, 4, 3	1, 2, 10	6, 12
7	18, 1	9, 6, 4	1, 2, 3, 4	18
8	9, 2	6, 4, 1	1, 2, 4, 6	9
9	9, 4	8, 3, 2	1, 2, 8	9
10	18, 4, 3	9, 8, 6, 2	1, 2, 8	18
11	9, 1	5, 3, 2	1, 2, 5	9
12	18, 5, 3	10, 9, 6, 1	1, 2, 10	18
13	18, 4	12, 9, 1	1, 3, 12	2, 18
14	12, 2	9, 4, 1	1, 9	2, 6, 12
15	18, 2	9, 6, 5	1, 3, 5	2, 18
16	10, 6	9, 5, 2	1, 9	2, 6, 10
17	14, 3	9, 7, 1	1, 9	2, 14
18	18, 3, 2	9, 7, 6, 1	1, 7	2, 18
19	12, 2	7, 4, 3	1, 7	2, 6, 12
20	14, 6, 4	12, 7, 3, 2	1, 3, 12	2, 14

Continued on next page

Index	a	b	K_-	K_+
21	14, 1	7, 5, 3	1, 3, 5	2, 14
22	10, 6, 1	7, 5, 3, 2	1, 7	2, 6, 10
23	15, 1	9, 5, 2	1, 2, 9	15
24	30, 9, 5	18, 15, 10, 1	1, 2, 18	30
25	15, 4	12, 5, 2	1, 2, 6, 12	15
26	30, 5, 4	15, 12, 10, 2	1, 2, 3, 12	30
27	15, 4	8, 6, 5	1, 2, 6, 8	15
28	30, 5, 4	15, 10, 8, 6	1, 2, 3, 8	30
29	15, 2	10, 4, 3	1, 2, 4, 10	15
30	30, 3, 2	15, 10, 6, 4	1, 2, 4, 5	30
31	30, 1	15, 10, 6	1, 2, 3, 5	30
32	15, 2	10, 6, 1	1, 2, 6, 10	15
33	15, 7	14, 5, 3	1, 2, 14	15
34	30, 5, 3	15, 10, 7, 6	1, 2, 7	30
35	30, 5, 3	15, 12, 10, 1	1, 2, 4, 12	30
36	15, 6, 1	12, 5, 3, 2	1, 2, 4, 12	15
37	15, 1	8, 5, 3	1, 2, 4, 8	15
38	30, 5, 3, 2	15, 10, 8, 6, 1	1, 2, 4, 8	30
39	20, 3	12, 10, 1	1, 2, 6, 12	20
40	20, 6, 1	12, 10, 3, 2	1, 2, 3, 12	20
41	20, 1	10, 8, 3	1, 2, 3, 8	20
42	20, 3, 2	10, 8, 6, 1	1, 2, 6, 8	20
43	20, 1	10, 7, 4	1, 2, 7	20
44	20, 7, 2	14, 10, 4, 1	1, 2, 14	20
45	20, 3	10, 9, 4	1, 2, 9	20
46	20, 9, 6	18, 10, 4, 3	1, 2, 18	20
47	24, 1	12, 8, 5	1, 2, 4, 5	24
48	24, 5, 2	12, 10, 8, 1	1, 2, 4, 10	24
49	24, 4, 1	12, 8, 7, 2	1, 2, 7	24
50	24, 7, 4	14, 12, 8, 1	1, 2, 14	24
51	24, 4, 3	12, 9, 8, 2	1, 2, 9	24
52	24, 9, 6, 4	18, 12, 8, 3, 2	1, 2, 18	24

B Reproducing the certificates and data

The implementation requires Python 3.10 or later and uses only the standard library. From the archive root, run

```
python code/verify_certificates.py
python code/test_research.py
```

The first command checks the supplied finite certificates without calling the generator. To regenerate all finite graphs, equality witnesses, sequence terms, and the atlas, run

```
python code/generate_artifacts.py
python code/verify_certificates.py
```

The checker deliberately raises an error if any state set is not closed, any transition violates its potential inequality, a required prime is missing, or a sharpness witness fails its direct valuation cal-

ulation. The large-prime assertion is supplied by the proved cutoff theorem and is not implemented as a search up to an arbitrary prime bound.

The central library also exposes `Ratio`, `Factor`, `classify`, `one_sided_capacity`, and `effective_bound`. For example:

```
from factorial_divisibility import A, Factor, classify, effective_bound

assert classify(A, (Factor(2, 1), Factor(3, 1), Factor(5, 1)))
assert not classify(A, (Factor(4, 1),))
assert not classify(A, (Factor(2, 1, multiplicity=2),))
cutoff, valid_multiplier = effective_bound(A, (Factor(2, 1),))
# The returned multiplier is a proved bound, not necessarily minimal.
```

When using the library from the archive root, add `code/` to `PYTHONPATH`, or run the example from within the code directory. Nonprimitive factors must be normalized and their common integer content treated separately. Equal roots must be merged by adding multiplicities before calling the classifier. This avoids silently mistaking two scalar multiples of a factor for two distinct roots.

The article compiles with ordinary pdfLaTeX and standard TeX Live packages. Run `make pdf`, or run `pdflatex article.tex` twice. The two atlas files and the certificate JSON are included, so compilation does not require a network connection or regeneration.

C Part II: an audit map of the results

Target	Result in this manuscript	Main proof location
A347854–A347858	All terms positive integral	Sections 13 and 14; exact residue floors
A347854–A347858	All five ordinary generating functions algebraic	Sections 17 and 18; Frobenius sampling and Gram certificates
A347855, June refinement	$3^n \mid a_2(n)$ for $3 \nmid n$	Section 15.1
A347854, June refinement	$(12n+1)\tilde{B}(n)/((2n+1)(6n+1))$ integral	Section 15.2
A295431, specified factors	All four multipliers 385, 5, 1, 770 work (re-proof; optimal by Part I)	Section 16.2; first proof Part I, Section 6
A295431, linear divisor	$U(n)/(12n-1)$ integral (re-proof)	Sections 15.2 and 16.3; first proof Part I, Proposition 5.3
A295431, $C(k, r)$	Explicit $(kL_{12r})^r$ for $k = 1, 2, 3$ (Part I's constant divides it)	Section 16.3; Part I, (26)
A295431, $D(r)$	Explicit L_r^r (Part I's constant divides it)	Section 16.3; Part I, Proposition 5.3
A295432	$6(6n+1)(12n-1) \mid \tilde{B}(n)$ for $n \geq 1$	Section 16.4
All five sequences	Compact positive moment laws; strict Hankel positivity	Section 19
All five sequences	All-orders expansions and inversion (instances of repository results, Section 10.3)	Sections 20 and 21

Target	Result in this manuscript	Main proof location
General family	Integral, algebraic rational dilation for arbitrary d	Theorem 12.1; Sections 13, 14 and 17.4

The OEIS status references were inspected for this manuscript on October 1, 2026. Their remaining conjecture labels are evidence for the target selection, not proof that no earlier unpublished or differently indexed proof exists. The mathematics above stands or falls on its arguments, not on a novelty label.

D Part II: package contents and rebuilding

The package includes `article.tex`, the compiled `article.pdf`, `verify.py`, its output `results.txt`, the generated `floor_certificates.json`, a table of the first 51 terms in `terms.csv`, the optional numerical demonstration, and a README. The TeX file is self-contained and uses no external image assets or bibliography processor. Rebuild it by running `pdflatex` twice. All essential mathematical certificate data appear in the manuscript itself.

[write] **Added 1 October 2026, batch 74: what is shipped.** The delivered manuscript, its PDF and its README are not shipped; this Part is the manuscript's text, built into this report's `article.tex`. The other delivered files are shipped with the prefix `02-fractional-dilation-`: the two programs in `code/`, the outputs `results.txt`, `floor_certificates.json`, `terms.csv`, `asymptotic_results.txt`, `asymptotic_results.csv` and `asymptotic_coefficients.csv` in `data/`, and `reviewer_notes.md` at the report root. The demonstration imports `verify` by its delivery name, and both programs write their outputs next to themselves under delivery names. To rerun them, copy the two programs into an empty scratch directory as `verify.py` and `asymptotic_demo.py`, run them there, and compare the outputs with the shipped `data/02-fractional-dilation-*` files; the report's README gives the commands. Do not run them from `code/`.

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